Analysis of imputation methods and missing indicators for Bayesian variable selection in clinical data

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Abstract

Background: Missing data complicates variable selection in clinical research, affecting both predictive performance and result interpretation. The impact of imputation methods on Bayesian variable selection and the utility of missing indicators require careful evaluation, particularly distinguishing prediction from inference goals.

Methods: We compared four imputation methods (MICE, mean, missForest, KNN) with Bayesian adaptive sampling (BAS) and Bayesian additive regression trees (BART) on two clinical datasets: MIMIC-III septic shock (64% missingness) and acute myocardial infarction outcomes (6.15% missingness). We also conducted simulation studies to evaluate coefficient bias, sensitivity, precision, and Type I and Type II errors under three missingness mechanisms (MCAR, MAR, MNAR).

Results: Missing indicators substantially improved predictive performance, particularly under high-missingness settings. Parsimonious models using 3–7 variables achieved best performance. Simulation revealed mechanism type dominated method choice: MAR caused large drops in predictive performance, while MCAR and MNAR had minimal impact on prediction. Mean imputation achieved competitive prediction but inflated Type I error (1.4–4 fold) and Type II error under MAR. Methods exhibited fundamental trade-offs between predictive performance and inferential validity.

Conclusions: Imputation method choice should align with analytical goals. For prediction, mean imputation with missing indicators can be optimal. For inference requiring unbiased estimates, MICE is essential. MAR mechanisms can

create unrecoverable information loss, emphasizing that preventing missingness is more important than correcting it statistically.

Keywords: Bayesian variable selection, missing data imputation, clinical data analysis, missing indicators, prediction versus inference, Type I and Type II error

1 Introduction

Variable selection is a critical step in developing statistical and machine learning models for clinical decision-making [1, 2]. By identifying informative predictors while excluding irrelevant ones, variable selection improves model interpretability, reduces overfitting, and enhances generalization [3].

Clinical datasets frequently contain substantial amounts of missing data due to equipment failure, patient dropout, and incomplete documentation. Such missingness can introduce bias, reduce statistical power, and compromise the reliability of variable selection. Without appropriate handling, it may lead to incorrect conclusions and degraded model performance [4–6].

Imputation methods are widely used to address missing data [4, 6–8]. Methods differ in their assumptions and ability to capture data structure. Multiple imputation by chained equations (MICE) generates multiple plausible values to reflect uncertainty [9–11]. Multiple imputation generally reduces bias compared to complete-case analysis, though method performance depends on the missingness mechanism [7]. Mean imputation replaces missing entries with observed averages [12]. missForest uses random forests for nonparametric estimation [13], though it may produce biased estimates compared to random forest implementations within MICE frameworks [14]. k-nearest neighbors (KNN) imputation estimates missing values from similar observations [15]. In addition, missing indicators, i.e. binary variables denoting whether a value is missing, can recover useful information under missing not at random (MNAR) mechanisms [16–18].

The influence of different imputation strategies on Bayesian variable selection has remained poorly understood. Existing work has either evaluated imputation methods in isolation [4, 6, 7, 14, 19] or examined traditional, non-Bayesian selection approaches [20, 21]. While [22] investigated imputation methods and missing indicators in clinical prediction models, their focus was on deployment scenarios and traditional prediction performance rather than Bayesian variable selection.

While recent studies have examined imputation methods in clinical prediction [11, 19, 22] and others have compared imputation approaches for missing data [4, 6–8, 14], our study made three distinct contributions. First, we systematically evaluated how imputation methods interact with Bayesian variable selection algorithms (BAS and BART), an area not addressed by prior work. Second, we explicitly tested the role of missing indicators across multiple imputation methods and missingness mechanisms, extending beyond previous evaluations that examined indicators in isolation [16–18]. Third, we distinguished between prediction-oriented and inference-oriented applications, providing context-specific recommendations rather than universal prescriptions.

Our work builds upon established findings that simple imputation methods can perform competitively for prediction tasks [11, 22] while acknowledging that multiple imputation remains essential for statistical inference [7, 23–25]. However, we extend these findings by demonstrating how method selection interacts with variable selection algorithms and missingness patterns, validated through both real clinical datasets and controlled simulations. This combined approach, using two datasets with contrasting characteristics (64% vs 6.15% missingness) alongside mechanism-specific simulations, enables us to identify when and why different strategies succeed or fail, providing actionable guidance for clinical researchers.

1.1 Prediction vs. inference in clinical modeling

Clinical data analysis serves two distinct goals with different methodological requirements: prediction and inference [26, 27]. Prediction tasks focus on accurate risk stratification for clinical decision support. The primary objective is maximizing predictive accuracy on new patients, typically measured by metrics such as AUC, calibration, and log-likelihood. Examples include predicting septic shock risk for ICU triage or identifying patients requiring intensive monitoring.

In contrast, inference tasks focus on unbiased estimation of treatment effects or risk factors, valid hypothesis testing, and accurate confidence intervals [26, 27]. These analyses aim to understand causal relationships between predictors and outcomes. Examples include estimating the effect of a specific intervention on mortality or identifying modifiable risk factors for clinical guidelines.

These goals have different implications for handling missing data [7, 26, 27]. For prediction-oriented applications, methods that maximize predictive accuracy are preferred, even if coefficient estimates are biased. For inference applications, methods that provide unbiased estimates and valid standard errors are essential, typically requiring multiple imputation frameworks with proper uncertainty quantification [4, 7, 11, 23, 25].

We used two clinical datasets with distinct missing data patterns: the MIMIC-III ICU database for septic shock prediction [19] and acute myocardial infarction (AMI) dataset on chronic heart failure outcomes. We evaluated four imputation methods: mean imputation [12], MICE [10, 11], missForest [13], and KNN [15], applied with Bayesian adaptive sampling (BAS) [28] for linear models and Bayesian additive regression trees (BART) [29] for nonlinear models. We employed Bayesian variable selection methods because they provide posterior inclusion probabilities for interpretable variable ranking and natural uncertainty quantification through model averaging, which frequentist methods like LASSO cannot provide without additional resampling procedures [28, 29]. We further assessed whether adding missing indicators improves predictive performance.

To evaluate variable selection performance and regression coefficients, we additionally conducted comprehensive simulation studies. The simulation design evaluated coefficient bias, Type I and Type II errors, sensitivity, and precision under three missingness mechanisms: missing completely at random (MCAR), missing at random

(MAR), and missing not at random (MNAR) [5, 24]. This combined approach, real data analysis for predictive evaluation and simulation for variable selection evaluation, provides both practical evidence and insights of when and why different imputation strategies succeed or fail.

Our goal was to identify optimal imputation strategies for Bayesian variable selection in prediction-oriented clinical applications with varying degrees of missingness. We demonstrated that the choice of imputation method depends on analytical goals: mean imputation excels for prediction tasks when combined with missing indicators, while MICE is essential for inference applications requiring unbiased coefficient estimates and proper uncertainty quantification.

2 Data and methods

2.1 Datasets and preprocessing

We used two clinical datasets with different missing data patterns.

MIMIC-III Dataset. The MIMIC-III dataset contains deidentified ICU data from Beth Israel Deaconess Medical Center (2001-2012) [30]. We extracted 168 ICU patients for binary classification of septic shock status (septic shock present versus absent). The dataset includes 52 covariates with 64% overall missingness (proportion of missing values across all covariate-observation pairs).

AMI Dataset. The AMI dataset examines chronic heart failure development following acute myocardial infarction [31, 32]. This dataset contains 1,699 observations with 120 covariates and 6.15% overall missingness. Binary outcomes: chronic heart failure versus no chronic heart failure.

Preprocessing. We standardized both datasets to zero mean and unit variance. Variables with more than 95% missing values were removed from both datasets prior to analysis, as they provide insufficient observed data for reliable imputation. For analyses including missing indicators, we created binary indicator variables defined in Section 2.4.1 and concatenated them with the imputed covariates to form augmented feature matrices, ensuring consistent covariate sets across all imputation methods.

2.2 Bayesian variable selection methods

2.2.1 Bayesian adaptive sampling (BAS)

Bayesian adaptive sampling (BAS) allows efficient exploration of the posterior for variable selection in logistic regression. We considered binary outcome variables $y_i \in \{0, 1\}$ for each observation $i = 1, \ldots, n$ and predictor matrix $X = [x_{ij}] \in \mathbb{R}^{n\times p}$, where x_{ij} represents predictor j for observation i, and p is the total number of predictors. The linear predictor for observation i is

$$z_i = \beta_0 + \sum_{j\in\mathcal{S}} \beta_j x_{ij}, \tag{1}$$

where $\mathcal{S} \subseteq \{1, \ldots, p\}$ represents selected variables, β_0 is the intercept, and β_j are the regression coefficients. We placed the following priors over model space $\mathcal{S}$ and coefficients

$$\begin{aligned}\beta_j \mid \mathcal{S} &\sim \mathcal{N}\left(0, g\cdot\left(X_{\mathcal{S}}^T X_{\mathcal{S}}\right)^{-1}\right),\\ P(\mathcal{S}) &\sim \text{Beta-Binomial}(1,1).\end{aligned} \tag{2}$$

We set hyperparameter $g = n$, following recommendations from [33] to balance model flexibility and regularization in high-dimensional settings. Here, $X_{\mathcal{S}} \in \mathbb{R}^{n\times|\mathcal{S}|}$ represents the submatrix of X containing the columns corresponding to selected predictors $\mathcal{S}$.

We used Markov chain Monte Carlo (MCMC) to estimate the marginal posterior inclusion probabilities

$$P(\beta_j \neq 0 \mid \mathbf{y}, X), \tag{3}$$

and selected the top-k predictors based on posterior inclusion probabilities.

We let $\beta_0^{(r)}$ and $\beta_j^{(r)}$ denote the r-th posterior samples. The posterior predictive probability is

$$\begin{aligned}P\left(\hat{y}_i = 1 \mid x_i^{\text{test}}, \mathbf{y}, X\right) &\approx \frac{1}{R}\sum_{r=1}^{R} P\left(\hat{y}_i = 1 \mid x_i^{\text{test}}, \beta^{(r)}\right),\\ &\approx \frac{1}{R}\sum_{r=1}^{R}\sigma\left(\beta_0^{(r)} + \sum_{j=1}^{k}\beta_j^{(r)} x_{ij}\right),\end{aligned} \tag{4}$$

where $\sigma(\cdot)$ is the logistic sigmoid function and R is the number of posterior samples.

To evaluate model performance, we used the average log predictive probability over the test set

$$\frac{1}{n^{\text{test}}}\sum_{i=1}^{n^{\text{test}}} \log P\left(\hat{y}_i = y_i^{\text{test}} \mid x_i^{\text{test}}, \mathbf{y}, X\right). \tag{5}$$

2.2.2 Bayesian additive regression trees (BART)

Bayesian additive regression trees (BART) models binary outcomes as nonlinear functions of p-dimensional predictor vectors $x = (x_1, x_2, \ldots, x_p)^T$. The probability of success is

$$\text{logit}(P(y = 1 \mid x)) = \sum_{m=1}^{M} g(x; T_m, \mu_m), \tag{6}$$

where $g(x; T_m, \mu_m)$ represents the m-th regression tree contribution, defined by binary tree structure T_m and terminal node parameters μ_m.

Each regression tree partitions the predictor space and assigns constant value $\mu_{m,\ell}$ to the ℓ-th terminal node. The model aggregates predictions from all M trees to compute log-odds. MCMC estimates the posterior distribution over trees.

For estimating variable inclusion probabilities, we used

$$\pi_j = \frac{1}{N\cdot M}\sum_{n=1}^{N}\sum_{m=1}^{M}\mathbb{I}(j \in T_m^{(n)}), \tag{7}$$

where $\mathbb{I}(\cdot)$ counts how often predictor j appears in splitting rules, $T_m^{(n)}$ is the m-th tree at MCMC iteration n, and N is total number of posterior samples.

We selected the top k variables with the highest π_j values. Predictive performance for both BAS and BART models was evaluated using Equation (5).

Implementation details for BAS and BART, including MCMC iterations, priors, and hyperparameters, are provided in Appendix B.

2.3 Imputation methods

We restricted our analysis to missing values in covariates only. The response variable y is fully observed in both datasets without any missing values. This assumption is common in clinical research where outcomes are typically recorded completely, while predictor measurements may be incomplete.

In the following subsections, we described the four imputation methods (missForest, KNN, MICE, and mean imputation) that we considered in our study.

2.3.1 missForest imputation

missForest uses iterative random forest imputation [13]. At iteration t, missing entries in variable x_j are imputed as:

$$x_j^{\text{mis},(t)} = \hat{f}_j^{(t)}(\mathbf{x}_{-j}), \tag{8}$$

where $\hat{f}_j^{(t)}$ represents the trained random forest model at iteration t and $\mathbf{x}_{-j}$ denotes all variables except the j-th one. The process continues until convergence or maximum iterations. This method handles mixed data types and captures complex nonlinear relationships. It assumes missing at random (MAR), which means that missingness can depend on the observed variables, but does not depend on the (unknown) missing value itself.

2.3.2 k-nearest neighbors (KNN) imputation

Let $x_{i\cdot}^{\text{mis}}$ denote the i-th observation with missing entries. We identified k-nearest neighbors $\mathcal{N}_i$ based on Euclidean distance computed on observed features [15]. For each missing variable x_{ij}, the imputed value is

$$x_{ij}^{\text{imp}} = \frac{1}{k} \sum_{\ell \in \mathcal{N}_i} x_{\ell j}^{\text{obs}}, \tag{9}$$

where $\mathcal{N}_i$ represents the index set of k nearest complete observations. KNN assumes similar instances have similar values for missing features under missing at random (MAR).

2.3.3 Multiple imputation by chained equations (MICE)

MICE treats each variable x_j with missing values as a response modeled conditionally on the remaining variables $\mathbf{x}_{-j}$ [10, 11]:

$$x_j^{\text{mis}} \sim P(x_j \mid \mathbf{x}_{-j}), \tag{10}$$

where $\mathbf{x}_{-j}$ denotes all variables except the j-th one. The algorithm iterates through variables with missing values, fitting a prediction model for each variable and imputing missing values based on that model.

We used predictive mean matching, where for each missing value x_{ij}^{mis}, we computed the predicted value $\hat{x}_{ij}$ from a fitted regression model. This prediction was compared to predicted values $\hat{x}_{kj}$ from observed cases. Finally, the imputed value is drawn from the observed value x_{kj}^{obs} with the closest prediction

$$x_{ij}^{\text{imp}} \leftarrow x_{kj}^{\text{obs}}, \quad \text{such that} \quad \hat{x}_{ij} \approx \hat{x}_{kj}. \tag{11}$$

MICE assumes missing at random (MAR) mechanism. The process generates m completed datasets:

$$\left\{X^{(1)}, X^{(2)}, \ldots, X^{(m)}\right\}. \tag{12}$$

For real datasets, we used $m = 20$ imputations to better capture uncertainty. For the simulation studies where we evaluated multiple conditions and mechanisms, we used $m = 3$ to balance computational efficiency with adequate uncertainty representation. For inference applications, results from the m datasets are combined using Rubin's rules to properly account for imputation uncertainty [23].

2.3.4 Mean imputation

For each variable x_j with missing entries, mean imputation replaces missing values with the sample mean of observed values [12]:

$$\bar{x}_j = \frac{1}{n_j^{\text{obs}}} \sum_{i: x_{ij} \text{ observed}} x_{ij}. \tag{13}$$

All missing values are replaced by:

$$x_{ij}^{\text{mis}} \leftarrow \bar{x}_j, \quad \forall i : x_{ij} \text{ is missing}. \tag{14}$$

This method assumes missing completely at random (MCAR) and does not incorporate uncertainty in imputed values.

Implementation details for all imputation methods are provided in Appendix B.

2.4 Missing indicators and experimental design

2.4.1 Definition and construction of missing indicators

Missing indicators are binary variables that explicitly encode the missingness pattern in the data. For each variable X_j with missing values, we created a corresponding indicator variable Z_j. For observation i and variable j, the missing indicator is defined as

$$Z_{i,j} = \begin{cases} 1 & \text{if } X_{i,j} \text{ is missing,} \\ 0 & \text{if } X_{i,j} \text{ is observed.} \end{cases} \tag{15}$$

Additionally, we computed the total number of missing values per observation

$$Z_{\text{total},i} = \sum_{j=1}^{p} Z_{i,j}. \tag{16}$$

After imputation, the augmented predictor matrix combines imputed values with missingness information

$$X_{\text{aug}} = [X_{\text{imp}} \mid Z], \tag{17}$$

where X_{imp} is the imputed data matrix and Z contains all non-constant missing indicators plus Z_{total}. This expands the feature space from p to approximately $2p$ variables, though the exact dimension depends on the number of constant indicators removed.

2.4.2 Role in the analysis pipeline

Missing indicators serve as additional predictors in the variable selection models. The analysis proceeded in two stages:

Stage 1 - Creating missing indicators and imputation: We created the missing indicator matrix Z before applying any imputation method. The imputation methods (mean, MICE, missForest, KNN) operated only on X to produce X_{imp}, filling missing entries with estimated values. The indicators Z remained unchanged and preserved the original missingness pattern.

Stage 2 - Modeling: Bayesian variable selection operates on the augmented matrix X_{aug}. Both BAS and BART select from the combined set of original variables and missing indicators. The final logistic regression model becomes

$$\text{logit}(P(y_i = 1)) = \beta_0 + \sum_{j \in \mathcal{S}_X} \beta_j X_{ij}^{\text{imp}} + \sum_{k \in \mathcal{S}_Z} \gamma_k Z_{ik}, \tag{18}$$

where $\mathcal{S}_X$ and $\mathcal{S}_Z$ are the selected sets of original variables and missing indicators, respectively.

Table 1 summarizes the final dataset after preprocessing and missing indicators construction.

2.4.3 Rationale for including missing indicators

Missing indicators can improve prediction under several mechanisms [16–18]. Under missing not at random (MNAR), missingness itself carries information about the unobserved values [17]. For example, extremely high or low glucose measurements may be missing due to critical patient status that precludes routine laboratory testing. The indicator Z_{glucose} captures this informative missingness even after X_{glucose} is imputed.

More generally, missingness patterns may correlate with unmeasured confounders [16, 22]. Missing laboratory values might indicate disease severity not fully captured by other recorded variables. By including Z_j as a predictor, the model can distinguish between observed and imputed values, effectively allowing different relationships for each group [18].

Additionally, all imputation methods introduce uncertainty. Even sophisticated approaches like MICE or missForest produce estimates rather than true values [4, 6]. Missing indicators alert the model that a value was imputed, potentially allowing a non-linear model like BART to weight observed values more heavily in prediction [22].

2.4.4 Experimental design

For each dataset (MIMIC-III and AMI), we evaluated two configurations:

Without missing indicators (w/o mi): Models use only imputed variables X_{imp} as predictors. **With missing indicators (w/ mi):** Models use the augmented matrix X_{aug} containing imputed variables, missing indicators, and total missing count. After preprocessing the datasets and removing constant indicators (variables with no missingness or complete missingness), the effective dimensions are shown in Table 1.

Table 1 Summary of datasets used for analysis

Dataset	Observations (n)	Vars w/o mi	Vars w/ mi
MIMIC-III (Septic Shock)	168	51	103
AMI (Chronic Heart Failure)	1,699	120	229

This table shows the number of observations and variables after all preprocessing steps. Original datasets contained 52 variables (MIMIC-III) and 121 variables (AMI). After removing the response variable, adding missing indicators, normalization, and removing constant or problematic features, the final dimensions shown above were used for analysis.

Abbreviations: Vars, Variables; mi, missing indicator; w/o, without; w/, with.

We used 10-fold cross-validation to evaluate performance. For each fold, missing indicators were derived from observed missingness patterns in the training set. Test set indicators were constructed independently from the test set's own missingness patterns, preventing data leakage.

Variable selection proceeds by ranking all available predictors. For BAS, we ranked by posterior inclusion probability $P(\beta_j \neq 0 \mid \mathbf{y}, X)$. For BART, we ranked by inclusion proportion π_j (Equation 7). We evaluated predictive performance for the top $k = 1, 2, \ldots, 20$ variables, constructing models with incrementally larger variable sets.

2.5 Evaluation metrics

We evaluated both prediction and variable selection performance using complementary metrics. For real clinical data, we focused on predictive performance measured by average log predicted probability, but also report AUC and F1 scores. For simulation studies, we additionally assessed coefficient estimation accuracy through relative bias and relative mean squared error, variable selection accuracy through sensitivity and precision, and error rate control through Type I and Type II error rates.

Performance was assessed using 10-fold cross-validation, with mean values and standard deviations reported across folds to demonstrate consistency. We focused on practical performance differences across methods rather than formal statistical significance testing. Given the large number of method-mechanism-dataset combinations and lack of a single primary comparison, formal hypothesis testing with multiple comparison corrections would be methodologically problematic and potentially misleading. Cross-validation provides empirical evidence of generalization performance, with consistency across folds indicating robustness.

2.5.1 Predictive performance metrics

Average log predictive probability (ALPP). We used ALPP for evaluating predictive performance, a proper scoring rule [22, 34] that assesses both predictive performance and calibration [35, 36] and was computed using Equation (5). ALPP has several advantages over threshold-dependent metrics [34]. As a proper scoring rule, it encourages models to output honest probability estimates rather than extreme predictions. It evaluates both predictive performance (the model's ability to rank observations correctly) and calibration (whether predicted probabilities match observed frequencies) [35, 36]. Unlike AUC, which only measures predictive performance through rank ordering [37], ALPP penalizes miscalibrated probabilities. Confident incorrect predictions receive larger penalties than uncertain incorrect predictions, reflecting clinical decision-making where confidence matters [38].

Values closer to zero indicate better performance. ALPP $= 0$ corresponds to perfect prediction, though this is impossible in practice. For example, ALPP $\approx$ -0.10 indicates average predicted probability of approximately 0.90 for the correct class, while ALPP $\approx$ -0.50 indicates average predicted probability of approximately 0.61. A difference of 0.05 in ALPP represents meaningful improvement in identifying high-risk patients [38].

Area under the ROC curve (AUC). AUC [37] measures predictive performance independent of threshold choice, ranging from 0.5 (random guessing) to 1.0 (perfect predictive performance). It quantifies the model's ability to rank positive cases higher than negative cases.

F1 score. Precision measures the proportion of predicted positives that are true positives, while recall (sensitivity) measures the proportion of actual positives correctly identified [39, 40]. These metrics are particularly relevant for imbalanced classification scenarios. F1 score [39, 40] is the harmonic mean of precision and recall:

$$\mathrm{F1} = 2 \cdot \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}. \tag{19}$$

It provides a single metric that balances both measures and is useful when both false positives and false negatives are important.

These threshold-dependent metrics complement ALPP in simulation studies by providing threshold-specific perspectives on model performance.

2.5.2 Variable selection metrics

In simulation studies where true variable effects are known, we evaluated variable selection performance using sensitivity, precision, Type I and Type II errors. We let $\mathcal{S}_{\text{true}} = \{j : \beta_j^{\text{true}} \neq 0\}$ denote the set of true variables and $\mathcal{S}_{\text{selected}}$ denote the selected variables.

Sensitivity (True Positive Rate): The proportion of true variables correctly identified:

$$\text{Sensitivity} = \frac{|\mathcal{S}_{\text{true}} \cap \mathcal{S}_{\text{selected}}|}{|\mathcal{S}_{\text{true}}|}. \tag{20}$$

Precision (Positive Predictive Value): The proportion of selected variables that are truly informative:

$$\text{Precision} = \frac{|\mathcal{S}_{\text{true}} \cap \mathcal{S}_{\text{selected}}|}{|\mathcal{S}_{\text{selected}}|}. \tag{21}$$

We reported precision rather than specificity because precision directly quantifies the reliability of the selected variable set, which is more informative in highly imbalanced scenarios where noise variables vastly outnumber signal variables [41].

2.6 Variable selection error rates

In variable selection, Type I and Type II errors measure selection accuracy rather than statistical significance, distinct from hypothesis testing frameworks where Type I error represents α (e.g., 0.05).

Type I Error (False Positive Rate): The proportion of noise variables ($\beta^{\text{true}} = 0$) incorrectly selected:

$$\text{Type I Error} = \frac{\text{FP}}{\text{FP} + \text{TN}}, \tag{22}$$

where FP is the number of noise variables incorrectly selected and TN is the number of noise variables correctly excluded.

Type II Error (False Negative Rate): The proportion of true predictors ($\beta^{\text{true}} \neq 0$) incorrectly excluded:

$$\text{Type II Error} = \frac{\text{FN}}{\text{FN} + \text{TP}}, \tag{23}$$

where FN is the number of true variables incorrectly excluded and TP is the number of true variables correctly selected.

Following standard practice in variable selection [42, 43], we used 5% as a reference threshold for Type I error, representing the expected false selection rate under well-calibrated methods, analogous to $\alpha = 0.05$ in hypothesis testing.

2.6.1 Coefficient estimation metrics

For simulation studies, we also evaluated the accuracy of coefficient estimates using relative bias and MSE. Relative bias quantifies systematic error for true variables with non-zero coefficients:

$$\text{Relative Bias} = \frac{1}{|\mathcal{S}_{\text{true}}|} \sum_{j \in \mathcal{S}_{\text{true}}} \frac{(\hat{\beta}_j - \beta_j^{\text{true}})}{\beta_j^{\text{true}}} \cdot 100\%, \tag{24}$$

where $\hat{\beta}_j$ is the estimated coefficient for true variable j, β_j^{true} is the known true coefficient, and $\mathcal{S}_{\text{true}}$ denotes the set of true variables.

Relative mean squared error (MSE) normalizes MSE by the squared true coefficients:

$$\text{Relative MSE} = \frac{\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^{\text{true}}\|_2^2}{\|\boldsymbol{\beta}^{\text{true}}\|_2^2} \cdot 100\%, \tag{25}$$

where $\hat{\boldsymbol{\beta}}$ contains estimated coefficients for the true variables, $\boldsymbol{\beta}^{\text{true}}$ are the corresponding true coefficients, and $\|\cdot\|_2$ denotes the L_2 norm.

2.7 Simulation study design

To estimate variable selection performance and the bias of regression coefficient estimates, we conducted several simulation studies. The simulation design mirrored the structure of both clinical datasets while allowing evaluation of coefficient estimation accuracy (relative bias and relative MSE), variable selection performance (sensitivity, precision, Type I and Type II errors), and prediction quality under controlled missingness mechanisms (MCAR, MAR, MNAR).

2.7.1 Data generation

For each dataset (MIMIC-III and AMI), we simulated the response y as follows:

Starting with missForest-imputed data $X_{\text{full}} \in \mathbb{R}^{n \times p}$, we preserved the actual covariate structure, distributions, and correlations present in the clinical datasets. We used missForest for this baseline imputation because its nonparametric approach captures complex relationships without parametric assumptions [13, 14], providing a realistic foundation while avoiding the selection bias that would result from complete-case analysis.

Step 1 - Missingness imposition: We imposed missingness under three mechanisms [4, 24] on X_{full} to create X_{missing}. For MCAR, we deleted values uniformly at random across all variables. For MCAR, we deleted values uniformly at random across all variables. For MAR, we partitioned variables into two groups: control variables (variables 1 to $p/2$) and dependent variables (variables $p/2 + 1$ to p). Missingness in dependent variables was created conditional on observed values in control variables, implementing a censoring mechanism where higher values in control variables increased the probability of missingness in dependent variables. For MNAR, we created self-censoring where the probability of a value being missing depended on that value itself, with higher values more likely to be missing. All three mechanisms were calibrated to match the observed missingness rates in real data (64% for MIMIC-III, 6.15% for AMI). Mathematical definitions of these mechanisms are provided in Appendix A.

Step 2 - True variable selection: From X_{missing}, we identified the four variables with highest missingness rates for each mechanism and assigned them as true predictors. This design ensures that true predictors have substantial missingness, avoiding scenarios where randomly selected variables are nearly complete and imputation methods have minimal opportunity to demonstrate differences. Coefficients were randomly assigned from $\boldsymbol{\beta}^{\text{true}} = \{1.5, 1.0, 0.5, 0.1\}$ without replacement. The remaining $p - 4$ variables were noise variables with $\beta_j^{\text{true}} = 0$ (48 noise variables for MIMIC-III, 117 for AMI). This design tests whether methods can recover variables that are both truly associated with the outcome and subject to substantial missingness, creating a realistic challenge for imputation and variable selection.

Step 3 - Outcome generation: Binary outcomes were generated via logistic regression using only the four true variables from X_{full} (the complete, non-missing values):

$$\begin{gathered} \text{logit}(P(y_i = 1 \mid X_i)) = \beta_0^{\text{true}} + \sum_{j \in \mathcal{S}_{\text{true}}} \beta_j^{\text{true}} X_{ij}, \\ y_i \sim \text{Bernoulli}(P(y_i = 1 \mid X_i)), \end{gathered} \tag{26}$$

where $\mathcal{S}_{\text{true}}$ denotes the four true variables. The intercept β_0^{true} was calibrated to match outcome prevalence in the real datasets: 35.7% positive cases for MIMIC-III (60 positives, 108 negatives) and 23.2% positive cases for AMI (394 positives, 1,305 negatives). This ensures simulated outcomes maintain realistic class balance across all three missingness mechanisms.

The simulation design described above addresses our primary research questions regarding the interaction between imputation methods and Bayesian variable selection. Additional sensitivity analyses examining alternative signal strengths ($c \in \{1, 2, 3\}$) applied directly to missing indicators are provided in Appendix F, which further explore the conditions under which missingness patterns themselves become predictive.

3 Results

3.1 Real data results

Figures 1 and 2 show BAS and BART performance across five different methods for both datasets. Missing indicators substantially improved performance, with effects most pronounced under high-missingness settings.

For MIMIC-III (64% missingness), including missing indicators substantially improved performance across all methods (Table 4). BAS with mean imputation showed the largest improvement when including missing indicators, with 92% relative improvement in average log posterior probability[1]. Other methods showed improvements ranging from 33% to 75% when including missing indicators. BART

[1]Relative improvement calculated as: $\frac{|\text{ALPP}_{\text{w/ mi}} - \text{ALPP}_{\text{w/o mi}}|}{|\text{ALPP}_{\text{w/o mi}}|} \times 100\%$. For ALPP (negative values), improvement means values closer to 0.

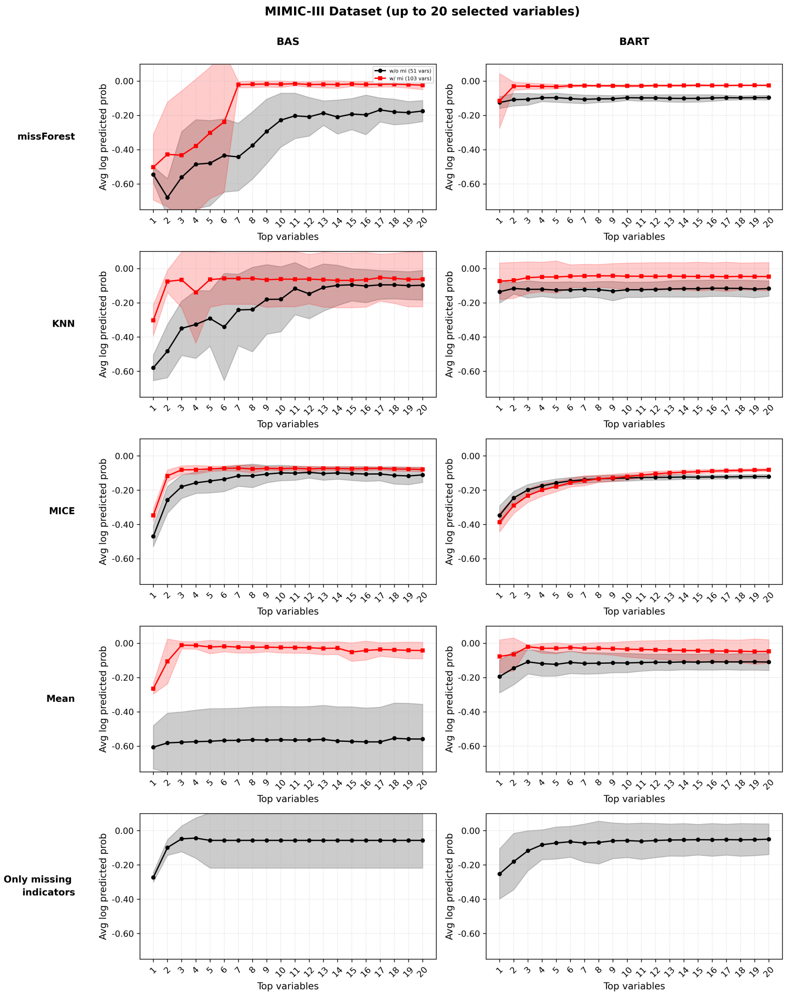

Fig. 1 Average log predicted probabilities from BAS (left) and BART (right) for MIMIC-III dataset across five imputation settings: (1) missForest, (2) KNN, (3) MICE (average across 20 imputations), (4) mean, and (5) missing indicators only. Shaded areas show standard deviation across folds. Red curves: with missing indicators; black curves: without missing indicators.

demonstrated similar patterns with smaller performance differences. Performance typically peaked within 3-7 variables across both models, supporting parsimonious variable selection.

For AMI (6.15% missingness), including missing indicators produced smaller but consistent improvements (Table 5). ALPP improvements ranged from 2% to 6% across methods, demonstrating that missing indicators provided meaningful predictive signal despite low overall missingness rates. Performance improvements occurred primarily within 5-10 variables.

The analyses above focus on variable selection up to 20 variables, which represents the primary scope of this study. For completeness, we also evaluated performance

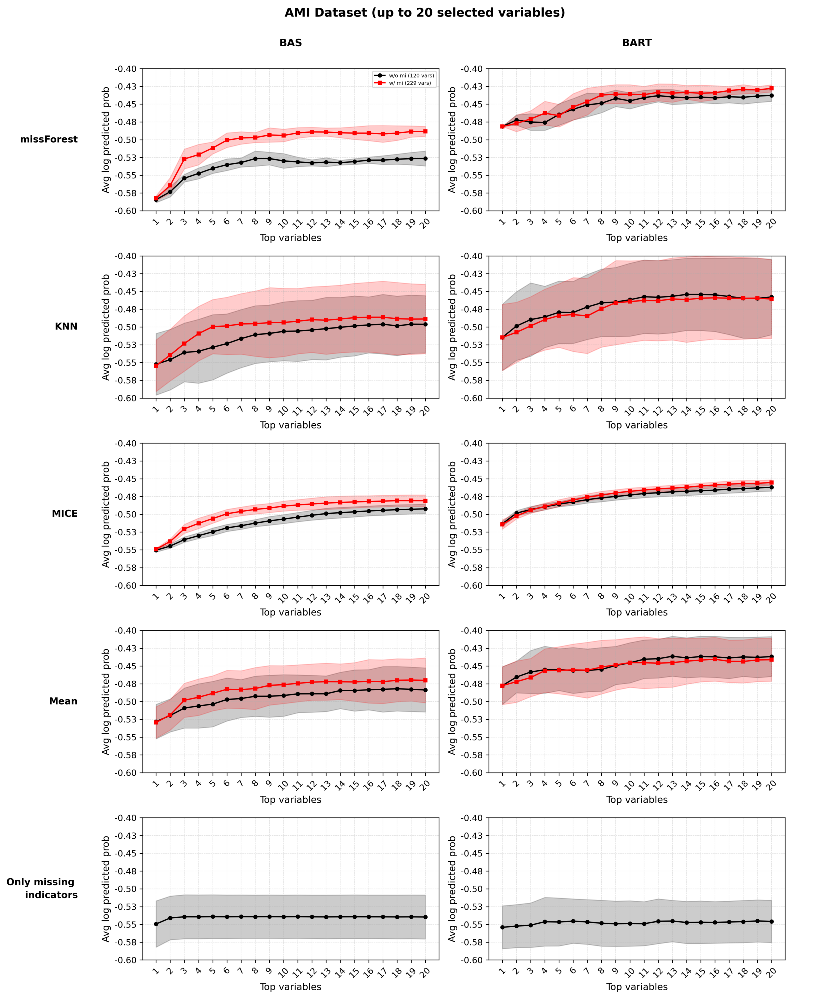

Fig. 2 Average log predicted probabilities from BAS (left) and BART (right) for AMI dataset across five imputation settings: (1) missForest, (2) KNN, (3) MICE (average across 20 imputations), (4) mean, and (5) missing indicators only. Shaded areas show standard deviation across folds. Red curves: with missing indicators; black curves: without missing indicators.

using the full variable set without variable selection (Appendices D and E). Consistent with our main findings, performance typically plateaued or declined beyond 20-30% of available variables across both datasets and methods, supporting the value of parsimonious variable selection for clinical applications.

Variable selection patterns are shown in Tables 2 and 3, with detailed descriptions below each table. Missing indicators consistently appeared among top-ranked variables, while BAS provided clearer prioritization than BART's distributed importance scores.

Table 2 Top Selected Variables by BAS and BART Across Imputation Methods for MIMIC-III Dataset

Model/Method	MIMIC-III
BAS-missForest	**Platelet Count (missing)** (0.894); **LD (missing)** (0.262); **Temperature** (0.257); Eosinophils (0.128); **O_2 Saturation** (0.123); pO_2 (0.057); **Lactate** (0.056); MCHC (missing) (0.055); **LD** (0.048); Urea Nitrogen (0.047)
BAS-KNN	**Platelet Count (missing)** (0.825); **O_2 Saturation** (0.108); **PT (missing)** (0.108); Base Excess (0.057); **Specific Gravity** (0.023); **RDW (missing)** (0.021); **ALT** (0.020); MCH (0.019); **LD** (0.015); Chloride (missing) (0.012)
BAS-MICE	**Platelet Count (missing)** (0.922); **LD (missing)** (0.480); Basophils (0.127); **LD** (0.095); Total missing (0.069); Lipase (missing) (0.067); Metamyelocytes (0.061); **Specific Gravity** (0.060); **Glucose** (0.059); Lactate (missing) (0.059)
BAS-Mean	**Platelet Count (missing)** (0.818); **PT (missing)** (0.115); **O_2 Saturation** (0.108); Base Excess (0.057); **Specific Gravity** (0.050); **RDW (missing)** (0.021); **ALT** (0.020); Chloride (missing) (0.012); Sodium (missing) (0.012); Atypical Lymphocytes (missing) (0.010)
BART-missForest	**Basophils (missing)** (0.014); Lipase (missing) (0.013); **Temperature (missing)** (0.012); **Lactate (missing)** (0.012); MCV (missing) (0.012); **Hemoglobin (missing)** (0.012); **Platelet Count (missing)** (0.012); pCO_2 (0.012); **Monocytes (missing)** (0.012); Atypical Lymphocytes (0.012)
BART-KNN	RBCs (missing) (0.013); Magnesium (missing) (0.012); pH (missing) (0.012); MCV (missing) (0.012); ALP (missing) (0.012); **Platelet Count (missing)** (0.012); **Basophils (missing)** (0.012); **LD (missing)** (0.011); **PT (missing)** (0.011); **Monocytes** (0.011)
BART-MICE	Creatinine (0.013); **PT** (0.013); ALP (0.013); CK (0.012); **Monocytes** (0.012); Phosphate (0.012); INR(PT) (0.012); pCO_2 (0.012); **Hemoglobin** (0.012); **Glucose** (0.012)
BART-Mean	Lymphocytes (missing) (0.014); Total missing (0.014); Sodium (missing) (0.014); Myelocytes (missing) (0.013); Metamyelocytes (missing) (0.012); **Temperature (missing)** (0.012); **PT (missing)** (0.012); **Hemoglobin (missing)** (0.012); **Platelet Count (missing)** (0.012); **LD (missing)** (0.011)

Top 10 variables selected by BAS and BART across four imputation methods for the MIMIC-III septic shock dataset. Values in parentheses indicate variable inclusion probabilities. Variables appearing consistently across methods include "platelet count", "lactate dehydrogenase (LD)", and "temperature." BAS shows clear prioritization with inclusion probabilities up to 0.922, while BART distributes importance more evenly with maximum probabilities of 0.014. Missing indicators (denoted with "missing" suffix) appear frequently among top-ranked variables. **Bold variables indicate those appearing in multiple methods or showing clinical relevance to septic shock outcomes.**

Abbreviations: BAS, Bayesian Adaptive Sampling; BART, Bayesian Additive Regression Trees; LD, Lactate Dehydrogenase; PT, Prothrombin Time; RDW, Red Cell Distribution Width; ALT, Alanine Aminotransferase; MCHC, Mean Corpuscular Hemoglobin Concentration; MCH, Mean Corpuscular Hemoglobin; MCV, Mean Corpuscular Volume; ALP, Alkaline Phosphatase; CK, Creatine Kinase; INR, International Normalized Ratio.

Table 3 Top Selected Variables by BAS and BART Across Imputation Methods for AMI Dataset

Model/Method	AMI
BAS-missForest	**NSAIDs ICU day 3 (missing)** (0.993); **Age** (0.941); **Chronic HF** (0.921); **Chronic bronchitis** (0.873); **Lateral MI** (0.864); **Opioid ICU day 3** (0.860); **SBP ICU (missing)** (0.567); **Atrial fib ECG** (0.565); **Pulmonary edema** (0.389); **Diabetes** (0.364)
BAS-KNN	**NSAIDs ICU day 3 (missing)** (0.993); **Age** (0.941); **Chronic HF** (0.921); **Chronic bronchitis** (0.873); **Lateral MI** (0.864); **Opioid ICU day 3** (0.654); **SBP ICU (missing)** (0.470); ECG: AF (0.435); **Pulmonary edema** (0.429); **Atrial fib ECG** (0.421)
BAS-MICE	**NSAIDs ICU day 3 (missing)** (0.993); **Age** (0.941); **Chronic HF** (0.921); **Chronic bronchitis** (0.873); Mobitz II (0.782); **Pulmonary edema** (0.698); **SBP ICU (missing)** (0.647); **Opioid ICU day 3** (0.470); Lidocaine ECG team (0.462); **Diabetes** (0.364)
BAS-Mean	**NSAIDs ICU day 3 (missing)** (0.993); **Age** (0.941); **Chronic HF** (0.921); **Chronic bronchitis** (0.873); **Lateral MI** (0.864); **Opioid ICU day 3** (0.860); **SBP ICU (missing)** (0.567); **Atrial fib ECG** (0.565); **Pulmonary edema** (0.389); **Diabetes** (0.364)
BART-missForest	**Chronic HF** (0.029); **NSAIDs ICU day 3 (missing)** (0.013); **Lateral MI** (0.012); **Chronic bronchitis** (0.011); **NSAIDs ECG team** (0.010); **Diabetes** (0.010); Pain relapse day 3 (missing) (0.010); **Lethal outcome** (0.009); Opioid ICU day 2 (missing) (0.009); **Opioid ICU day 3 (missing)** (0.009)
BART-KNN	**Chronic HF** (0.031); **NSAIDs ICU day 3 (missing)** (0.013); **Chronic bronchitis** (0.011); **Age** (0.011); **Opioid ICU day 3 (missing)** (0.010); **Diabetes** (0.009); **Lateral MI** (0.009); **Lethal outcome** (0.009); **NSAIDs ECG team** (0.009); **Pulmonary edema** (0.009)
BART-MICE	**Chronic HF** (0.027); **NSAIDs ICU day 3 (missing)** (0.012); **Age** (0.011); **Chronic bronchitis** (0.010); **Opioid ICU day 3 (missing)** (0.010); Opioid ICU day 2 (missing) (0.010); **Pulmonary edema** (0.009); **Diabetes** (0.009); **Lethal outcome** (0.008); Lidocaine ECG team (0.008)
BART-Mean	**Chronic HF** (0.032); **Chronic bronchitis** (0.012); **Opioid ICU day 3 (missing)** (0.011); **NSAIDs ICU day 3 (missing)** (0.011); **Age** (0.011); **Lethal outcome** (0.010); **Pulmonary edema** (0.009); **Lateral MI** (0.009); **Diabetes** (0.009); Dressler syndrome (0.009)

Top 10 variables selected by BAS and BART across four imputation methods for the AMI dataset. Values in parentheses indicate variable inclusion probabilities. Variables appearing consistently across methods include chronic conditions ("chronic heart failure", "chronic bronchitis", "diabetes"), medications ("NSAIDs", "opioids"), and cardiac events ("atrial fibrillation", "pulmonary edema", "lateral MI"). BAS demonstrates strong variable prioritization with inclusion probabilities up to 0.993, while BART shows more distributed importance with maximum probabilities of 0.032. Missing indicators such as "NSAIDs ICU day 3 (missing)" and "SBP ICU (missing)" consistently appear among top-ranked variables despite lower overall missingness rates (6.15%), providing predictive information about outcomes. **Bold variables indicate those appearing in multiple methods or showing clinical relevance to MI outcomes.**

Abbreviations: BAS, Bayesian Adaptive Sampling; BART, Bayesian Additive Regression Trees; NSAIDs, Non-Steroidal Anti-Inflammatory Drugs; ICU, Intensive Care Unit; HF, Heart Failure; MI, Myocardial Infarction; SBP, Systolic Blood Pressure; ECG, Electrocardiogram; AF, Atrial Fibrillation.

3.1.1 BAS performance

Tables 4 and 5 present BAS performance across imputation methods.

Table 4 Bayesian Adaptive Sampling (BAS) Performance for real MIMIC-III Dataset: Variable Selection Results Across Imputation Methods

Method	mi	AUC	F1 Score	ALPP
KNN	w/o mi	0.9932 (0.0216)	0.8864 (0.1156)	-0.1442 (0.0689)
	w/ mi	0.9938 (0.0198)	0.9654 (0.0460)	-0.0853 (0.1575)
MICE	w/o mi	0.9921 (0.0328)	0.9645 (0.0688)	-0.0984 (0.1286)
	w/ mi	0.9941 (0.0207)	0.9700 (0.0887)	-0.0722 (0.1522)
Mean	w/o mi	0.6905 (0.1841)	0.3440 (0.3730)	-0.5549 (0.1941)
	w/ mi	**1.0000 (0.0000)**	**0.9766 (0.0508)**	**-0.0242 (0.0312)**
missForest	w/o mi	0.9932 (0.0111)	0.9041 (0.0761)	-0.1419 (0.0733)
	w/ mi	**1.0000 (0.0000)**	0.9446 (0.1037)	-0.0511 (0.0629)

Bayesian Adaptive Sampling (BAS) performance for four imputation methods (MICE, Mean, missForest, KNN) with and without missing indicators (w/ mi, w/o mi) on the MIMIC-III dataset. Metrics include: AUC (area under the ROC curve for predictive performance), F1 Score (harmonic mean of precision and recall), and ALPP (average log posterior probability). Values are presented as mean with standard deviation (SD) in parentheses across 10-fold cross-validation. **Bold values indicate best performance for each metric.**

Abbreviations: mi, missing indicator; BAS, Bayesian Adaptive Sampling; w/ mi, with missing indicators; w/o mi, without missing indicators; AUC, Area Under the ROC Curve; F1, F1 Score; ALPP, Average Log Posterior Probability.

For MIMIC-III, missing indicators transformed mean imputation performance, achieving perfect AUC (45% improvement over mean alone) and best calibration. Without indicators, sophisticated methods (MICE, KNN, missForest) substantially outperformed mean imputation (44% better). With indicators, this gap disappeared completely.

For AMI, including missing indicators produced modest (2-5% AUC gains) but consistent improvements. Mean with indicators achieved best F1 (27% improvement) and calibration, while KNN with indicators showed best predictive performance (7% better than mean without indicators).

Missing indicator benefits increased with missingness rate: dramatic for MIMIC-III (64%), modest for AMI (6.15%). These patterns suggest missingness patterns themselves carry prognostic information in high-missingness settings.

3.2 Results on simulated data

Here we report the results for simulation using the design described in Section 2.7.

3.2.1 Coefficient recovery

Figures 3 and 4 show coefficient recovery across four true effect sizes ($\beta = 0.1, 0.5, 1.0, 1.5$), comparing models with missing indicators (red) versus without (black) under three missingness mechanisms.

Table 5 Bayesian Adaptive Sampling (BAS) Performance for real AMI Dataset: Variable Selection Results Across Imputation Methods

Method	**mi**	**AUC**	**F1 Score**	**ALPP**
KNN	w/o mi	0.7471 (0.0378)	0.2492 (0.0472)	-0.4881 (0.0461)
	w/ mi	**0.7656 (0.0188)**	0.2575 (0.0838)	-0.4710 (0.0397)
MICE	w/o mi	0.7378 (0.0474)	0.2569 (0.0901)	-0.4875 (0.0437)
	w/ mi	0.7607 (0.0395)	0.2790 (0.0888)	-0.4688 (0.0408)
Mean	w/o mi	0.7185 (0.0507)	0.2245 (0.0855)	-0.4902 (0.0327)
	w/ mi	0.7567 (0.0683)	**0.2855 (0.1154)**	**-0.4662 (0.0574)**
missForest	w/o mi	0.7386 (0.0388)	0.2626 (0.0694)	-0.4946 (0.0464)
	w/ mi	0.7600 (0.0261)	0.2772 (0.0940)	-0.4763 (0.0232)

Bayesian Adaptive Sampling (BAS) performance for four imputation methods (MICE, Mean, missForest, KNN) with and without missing indicators (w/ mi, w/o mi) on the AMI dataset. Metrics include: AUC (area under the ROC curve for predictive performance), F1 Score (harmonic mean of precision and recall), and ALPP (average log posterior probability). Values are presented as mean with standard deviation (SD) in parentheses across 10-fold cross-validation. **Bold values indicate best performance for each metric.**

Abbreviations: mi, missing indicator; BAS, Bayesian Adaptive Sampling; w/ mi, with missing indicators; w/o mi, without missing indicators; AUC, Area Under the ROC Curve; F1, F1 Score; ALPP, Average Log Posterior Probability..

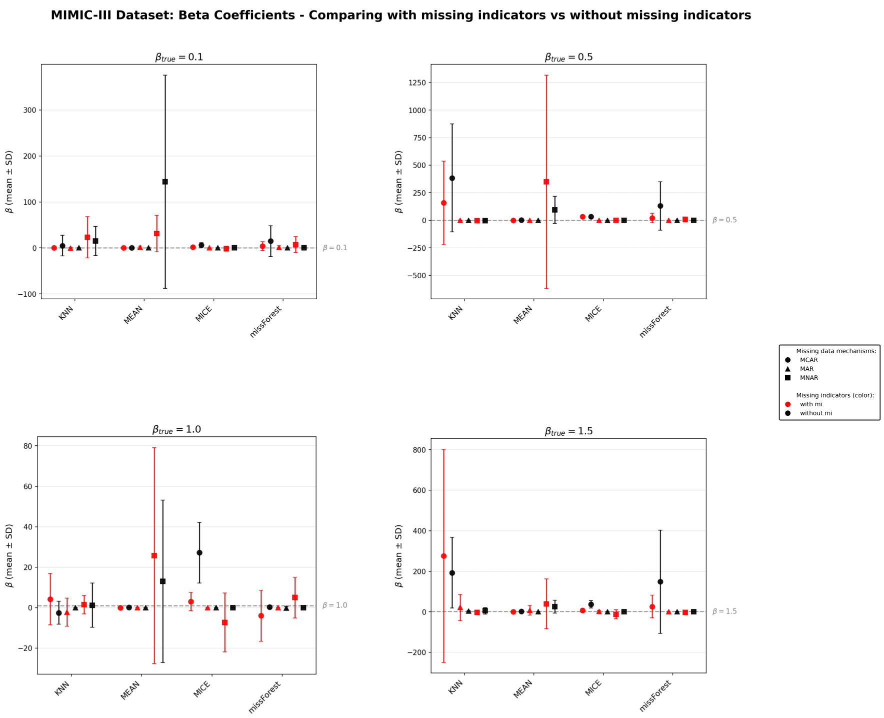

Fig. 3 Coefficient recovery for MIMIC-III dataset comparing models with missing indicators (red) versus without missing indicators (black) across four true effect sizes ($\beta = 0.1, 0.5, 1.0, 1.5$). Each panel shows mean predicted coefficients ± SD across 10-fold cross-validation for three missingness mechanisms (circles: MCAR, triangles: MAR, squares: MNAR) and four imputation methods (KNN, MEAN, MICE, missForest). Dashed horizontal line indicates true coefficient value. Missing indicators improve recovery under MNAR but provide minimal benefit under MCAR and cannot recover MAR failures.

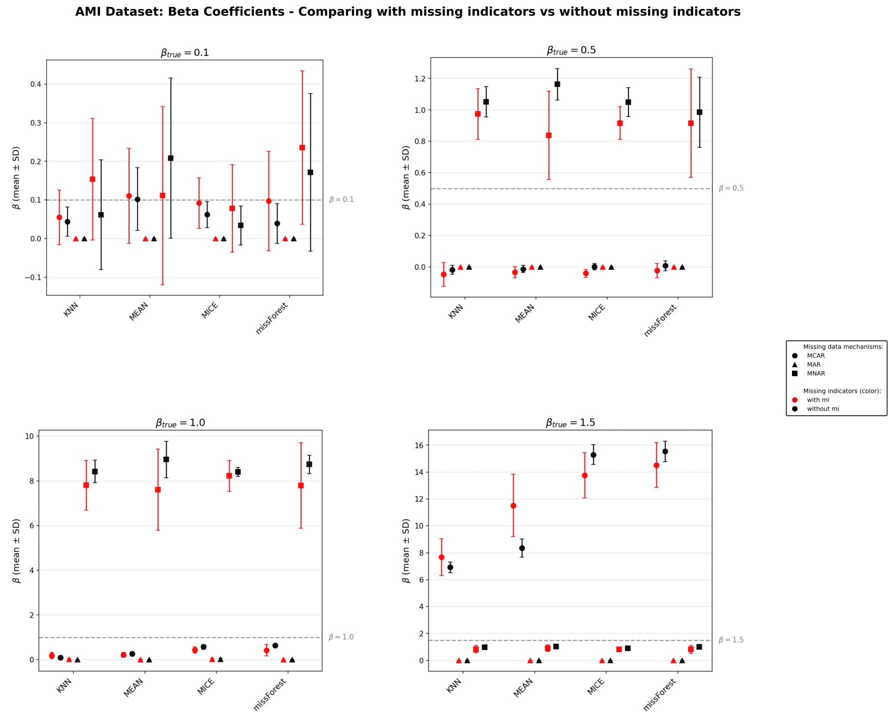

Fig. 4 Coefficient recovery for AMI dataset comparing models with missing indicators (red) versus without missing indicators (black) across four true effect sizes ($\beta = 0.1, 0.5, 1.0, 1.5$). Each panel shows mean predicted coefficients ± SD across 10-fold cross-validation for three missingness mechanisms (circles: MCAR, triangles: MAR, squares: MNAR) and four imputation methods (KNN, MEAN, MICE, missForest). Dashed horizontal line indicates true coefficient value. Missing indicators substantially improve MNAR recovery, show minimal MCAR impact, and cannot overcome MAR information loss.

Under MCAR, missing indicators showed minimal impact on coefficient recovery, with both conditions achieving similar performance. Under MAR, all methods failed completely regardless of indicator inclusion, with coefficients collapsing to near-zero. Under MNAR, missing indicators substantially improved recovery across all methods, demonstrating their value when missingness depends on unobserved values. Comprehensive performance metrics including sensitivity, precision, and error rates are presented in Tables 6 and 7.

3.2.2 Comprehensive performance evaluation

Tables 6 and 7 present comprehensive results across all conditions.

For MIMIC-III, under MCAR produced strongest performance with MICE achieving best classification but poor coefficient recovery. Mean imputation showed 99.9% better relative MSE than MICE despite weaker prediction, demonstrating prediction-inference trade-offs. MAR caused severe degradation (20% lower AUC) with coefficient collapse. Missing indicators provided minimal under MNAR benefit for MIMIC-III but substantial benefit for AMI.

Table 6 Comprehensive Simulation Results for MIMIC-III Dataset by Missingness Mechanism, Missing Indicator Condition, and Imputation Method

			Classification Performance			Coefficient Recovery			
Mechanism	**mi**	**Method**	**AUC**	**F1**	**ALPP**	**Sensitivity**	**Precision**	**Rel MSE**	**Rel Bias**
MAR	w/o mi	KNN	0.6174 (0.1133)	**0.2285 (0.2153)**	**-0.6291 (0.0717)**	**0.550 (0.197)**	**0.110 (0.039)**	300.50 (312.51)	**-37.44 (36.13)**
		Mean	**0.6301 (0.1256)**	0.1555 (0.1684)	-0.6556 (0.0866)	0.475 (0.249)	0.095 (0.050)	**88.53 (8.59)**	-97.24 (2.17)
		MICE	0.6293 (0.0599)	**0.2285 (0.1656)**	-0.6549 (0.0795)	0.475 (0.249)	0.095 (0.050)	98.31 (5.63)	-101.38 (1.11)
		missForest	0.5362 (0.1074)	0.1143 (0.1475)	-0.6611 (0.0621)	0.250 (0.289)	0.050 (0.058)	130.62 (96.66)	-102.44 (7.85)
	w/ mi	KNN	0.6420 (0.1058)	0.2013 (0.1864)	-0.6573 (0.0607)	**0.400 (0.211)**	0.113 (0.065)	117941.77 (372405.13)	136.32 (681.48)
		Mean	**0.6802 (0.0936)**	**0.2802 (0.2518)**	**-0.6567 (0.1105)**	0.375 (0.243)	**0.116 (0.074)**	16632.47 (52321.62)	252.08 (1104.16)
		MICE	0.5684 (0.0827)	0.1003 (0.1402)	-0.6692 (0.0681)	0.200 (0.258)	0.053 (0.064)	764.68 (2119.31)	**-74.12 (90.33)**
		missForest	0.5307 (0.1416)	0.0786 (0.1732)	-0.6664 (0.0774)	0.175 (0.313)	0.047 (0.084)	**497.85 (1257.86)**	197.68 (941.88)
MCAR	w/o mi	KNN	0.5513 (0.0889)	0.1483 (0.2297)	-0.6447 (0.0890)	**0.600 (0.175)**	0.120 (0.035)	12186935.59 (19901947.53)	23536.25 (23614.69)
		Mean	0.7399 (0.1272)	0.2773 (0.2239)	-0.6207 (0.1318)	0.525 (0.079)	0.105 (0.016)	**116.05 (105.05)**	**19.85 (51.66)**
		MICE	**0.8462 (0.0651)**	**0.6507 (0.1101)**	**-0.5292 (0.1516)**	0.550 (0.197)	0.110 (0.039)	106577.31 (103329.18)	4420.60 (2082.67)
		missForest	0.6864 (0.1178)	0.3409 (0.2112)	-0.6255 (0.1491)	0.675 (0.237)	**0.135 (0.047)**	4023526.76 (7595935.15)	12571.20 (22562.21)
	w/ mi	KNN	0.5094 (0.0665)	0.0682 (0.1113)	-0.6468 (0.0750)	0.500 (0.118)	**0.166 (0.036)**	13675398.83 (24118113.28)	12574.46 (20151.63)
		Mean	0.7100 (0.1054)	0.3245 (0.2106)	-0.6217 (0.0888)	0.500 (0.204)	0.160 (0.058)	**59.70 (19.91)**	**-45.51 (48.07)**
		MICE	**0.7718 (0.0677)**	**0.5338 (0.0934)**	**-0.5818 (0.1337)**	0.525 (0.219)	0.128 (0.055)	38832.95 (35092.57)	2228.97 (1086.85)
		missForest	0.6132 (0.1094)	0.1686 (0.2255)	-0.6513 (0.0994)	**0.600 (0.394)**	0.176 (0.119)	164777.77 (347913.10)	2295.94 (5064.20)
MNAR	w/o mi	KNN	0.6580 (0.1500)	0.2236 (0.2039)	**-0.6222 (0.1305)**	0.675 (0.265)	0.135 (0.053)	45304.94 (98355.58)	3656.46 (7800.91)
		Mean	0.6385 (0.1298)	**0.3670 (0.2398)**	-0.6507 (0.1390)	**0.900 (0.129)**	**0.180 (0.026)**	2698192.08 (6683522.71)	41416.90 (63870.32)
		MICE	0.5678 (0.0807)	0.0080 (0.0253)	-0.6811 (0.0817)	0.675 (0.290)	0.135 (0.058)	**99.50 (0.50)**	**-99.22 (0.83)**
		missForest	**0.6700 (0.1309)**	0.1400 (0.2989)	-0.6390 (0.1190)	0.700 (0.307)	0.140 (0.061)	99.75 (0.31)	-98.75 (2.17)
	w/ mi	KNN	**0.5881 (0.1257)**	0.0722 (0.1657)	**-0.6489 (0.0827)**	0.500 (0.312)	0.195 (0.083)	**72350.62 (149715.73)**	5604.99 (11213.05)
		Mean	0.5537 (0.1505)	**0.1108 (0.1521)**	-0.6648 (0.0834)	0.500 (0.167)	**0.222 (0.090)**	28076727.91 (85827694.02)	26471.00 (47123.73)
		MICE	0.5674 (0.1053)	0.0722 (0.1657)	-0.6570 (0.0767)	0.400 (0.316)	0.216 (0.190)	27295.28 (45979.91)	**-884.52 (1720.51)**
		missForest	**0.5881 (0.1257)**	0.0722 (0.1657)	**-0.6489 (0.0827)**	**0.475 (0.299)**	0.188 (0.110)	28553.86 (61473.12)	2149.16 (4790.55)

Results for the MIMIC-III dataset, showing performance across three missingness mechanisms (MCAR, MAR, MNAR), two missing indicator conditions (w/o mi, w/ mi), and four imputation methods (KNN, MEAN, MICE, missForest). Classification Performance: AUC (area under ROC curve for predictive performance), F1 Score (harmonic mean of precision and recall), log-likelihood. Coefficient Recovery: Sensitivity (true positive rate), Precision (positive predictive value), Relative MSE and Relative Bias compared to complete data. Values presented as mean with standard deviation in parentheses across simulation replications. **Bold values indicate best performance for each metric within each mechanism-mi combination.**

Abbreviations: mi, missing indicator; MCAR, Missing Completely At Random; MAR, Missing At Random; MNAR, Missing Not At Random; w/ mi, with missing indicators; w/o mi, without missing indicators; KNN, *k*-Nearest Neighbors; MEAN, Mean Imputation; MICE, Multivariate Imputation by Chained Equations; AUC, Area Under the ROC Curve; F1, F1 Score; ALPP, Log-Likelihood; Rel MSE, Relative Mean Squared Error; Rel Bias, Relative Bias.

Table 7 Comprehensive Simulation Results for AMI Dataset by Missingness Mechanism, Missing Indicator Condition, and Imputation Method

			Classification Performance			Coefficient Recovery			
Mechanism	**mi**	**Method**	**AUC**	**F1**	**ALPP**	**Sensitivity**	**Precision**	**Rel MSE**	**Rel Bias**
MAR	w/o mi	KNN	**0.5348 (0.0439)**	0.0000 (0.0000)	**-0.5682 (0.0265)**	**0.050 (0.158)**	**0.010 (0.032)**	99.87 (0.43)	-99.95 (0.17)
		Mean	**0.5348 (0.0439)**	0.0000 (0.0000)	**-0.5682 (0.0265)**	**0.050 (0.158)**	**0.010 (0.032)**	99.88 (0.37)	-99.95 (0.15)
		MICE	0.5212 (0.0340)	**0.0015 (0.0048)**	-0.5683 (0.0266)	0.000 (0.000)	0.000 (0.000)	**98.90 (2.52)**	**-99.49 (1.18)**
		missForest	0.5296 (0.0430)	0.0000 (0.0000)	-0.5683 (0.0265)	**0.050 (0.158)**	**0.010 (0.032)**	99.96 (0.14)	-99.99 (0.04)
	w/ mi	KNN	**0.5368 (0.0416)**	0.0000 (0.0000)	-0.5684 (0.0266)	**0.050 (0.105)**	0.010 (0.021)	99.24 (1.84)	-99.65 (0.85)
		Mean	0.5247 (0.0385)	0.0000 (0.0000)	-0.5684 (0.0267)	**0.050 (0.105)**	**0.011 (0.022)**	99.73 (0.59)	-99.88 (0.26)
		MICE	0.5321 (0.0223)	**0.0027 (0.0058)**	**-0.5683 (0.0262)**	0.075 (0.169)	0.018 (0.041)	**99.14 (2.25)**	**-99.60 (1.06)**
		missForest	0.5366 (0.0388)	0.0000 (0.0000)	-0.5685 (0.0266)	0.025 (0.079)	0.006 (0.020)	99.96 (0.12)	-99.98 (0.05)
MCAR	w/o mi	KNN	0.9878 (0.0088)	0.9469 (0.0112)	-0.1374 (0.0274)	**0.950 (0.105)**	**0.190 (0.021)**	**872.35 (121.22)**	**27.69 (8.16)**
		Mean	0.9895 (0.0051)	0.9433 (0.0164)	-0.1220 (0.0186)	0.875 (0.132)	0.175 (0.026)	1373.15 (238.69)	70.71 (20.29)
		MICE	**0.9964 (0.0017)**	0.9550 (0.0141)	-0.0771 (0.0135)	0.925 (0.121)	0.185 (0.024)	5448.66 (572.18)	184.85 (14.42)
		missForest	0.9971 (0.0010)	**0.9563 (0.0111)**	**-0.0716 (0.0111)**	0.900 (0.129)	0.180 (0.026)	5635.02 (614.05)	184.93 (17.97)
	w/ mi	KNN	0.9899 (0.0054)	0.9310 (0.0204)	-0.1230 (0.0266)	0.800 (0.158)	**0.261 (0.048)**	**1168.06 (420.26)**	**44.26 (15.90)**
		Mean	0.9932 (0.0021)	0.9372 (0.0141)	-0.1046 (0.0128)	0.875 (0.177)	0.260 (0.039)	3021.23 (1262.91)	123.35 (49.61)
		MICE	**0.9962 (0.0018)**	**0.9531 (0.0146)**	-0.0799 (0.0147)	**1.000 (0.000)**	0.259 (0.033)	4376.18 (1136.64)	161.38 (22.13)
		missForest	0.9968 (0.0014)	0.9555 (0.0124)	**-0.0739 (0.0121)**	0.800 (0.158)	0.231 (0.053)	4919.30 (1193.67)	175.79 (36.50)
MNAR	w/o mi	KNN	0.9933 (0.0030)	0.9220 (0.0232)	-0.0906 (0.0197)	0.925 (0.121)	0.185 (0.024)	1591.28 (205.56)	194.90 (41.49)
		Mean	**0.9941 (0.0023)**	0.9201 (0.0327)	**-0.0892 (0.0173)**	**1.000 (0.000)**	**0.200 (0.000)**	1840.36 (353.22)	251.33 (64.62)
		MICE	0.9928 (0.0029)	0.9142 (0.0236)	-0.0947 (0.0195)	0.975 (0.079)	0.195 (0.016)	**1578.04 (84.43)**	**185.65 (13.74)**
		missForest	0.9935 (0.0030)	**0.9250 (0.0226)**	-0.0912 (0.0192)	**1.000 (0.000)**	**0.200 (0.000)**	1725.91 (179.49)	226.88 (63.45)
	w/ mi	KNN	**0.9941 (0.0027)**	0.9201 (0.0250)	-0.0877 (0.0195)	0.975 (0.079)	**0.381 (0.082)**	1370.43 (395.52)	**174.99 (101.96)**
		Mean	0.9939 (0.0026)	0.9158 (0.0259)	-0.0907 (0.0190)	0.925 (0.121)	0.378 (0.166)	**1345.49 (544.62)**	195.93 (62.57)
		MICE	0.9933 (0.0028)	0.9180 (0.0235)	-0.0918 (0.0191)	0.975 (0.079)	0.342 (0.082)	1518.33 (269.06)	184.73 (31.34)
		missForest	0.9942 (0.0022)	**0.9250 (0.0226)**	**-0.0876 (0.0161)**	0.975 (0.079)	0.364 (0.103)	1432.63 (572.31)	212.99 (105.50)

Results for the AMI dataset, showing performance across three missingness mechanisms (MCAR, MAR, MNAR), two missing indicator conditions (w/o mi, w/ mi), and four imputation methods (KNN, MEAN, MICE, missForest). Classification Performance: AUC (area under ROC curve for predictive performance), F1 Score (harmonic mean of precision and recall), log-likelihood. Coefficient Recovery: Sensitivity (true positive rate), Precision (positive predictive value), Relative MSE and Relative Bias compared to complete data. Values presented as mean with standard deviation in parentheses across simulation replications. **Bold values indicate best performance for each metric within each mechanism-mi combination.**

Abbreviations: mi, missing indicators; MCAR, Missing Completely At Random; MAR, Missing At Random; MNAR, Missing Not At Random; w/ mi, with missing indicators; w/o mi, without missing indicators; KNN, k-Nearest Neighbors; MEAN, Mean Imputation; MICE, Multivariate Imputation by Chained Equations; AUC, Area Under the ROC Curve; F1, F1 Score; ALPP, Log-Likelihood; Rel MSE, Relative Mean Squared Error; Rel Bias, Relative Bias.

For AMI, under MCAR produced near-perfect classification but coefficient inflation exceeding 770% even for best methods. MAR caused complete failure (47% worse AUC, sensitivity dropping from 80-100% to 0-7.5%). MNAR restored strong performance with missing indicators improving precision by 90%. Mean imputation with indicators achieved best coefficient recovery under MNAR (15% better relative MSE).

Cross-dataset patterns: (1) Mechanism effects dominate method choice: MCAR and MNAR produce minimal degradation in predictive performance, while MAR causes large drops across all methods. (2) Prediction-inference trade-offs are pervasive: methods optimized for prediction show inflated coefficients. (3) Missing indicators provide substantial improvements under MNAR but minimal benefit under MCAR, with all methods showing degraded performance under MAR. (4) MIMIC's low sample-to-feature ratio ($n/p = 3.2$) results in high variability in coefficient estimates (large standard deviations) compared to AMI ($n/p = 14.0$).

3.2.3 Sensitivity Analysis: Amplified Missingness Effect ($\alpha = 10$)

To investigate performance when outcomes depend solely on missingness patterns, we conducted a sensitivity analysis with amplified coefficients ($\alpha = 10$), setting predictor coefficients to zero while increasing missing indicator coefficients from $\{0.1, 0.5, 1.0, 1.5\}$ to $\{1.0, 5.0, 10.0, 15.0\}$. Results are presented in Table 8.

Signal amplification produced substantial improvements across all metrics for both datasets. For AMI, MCAR achieved strongest performance (AUC up to 0.967). For MIMIC, MNAR using missForest achieved near-perfect classification (AUC 0.995).

This contrasts sharply with main simulation results (Tables 6, 7) where outcomes depended on predictor values. Under standard simulation, MAR caused universal failure regardless of method, while MCAR enabled strong performance without indicators. The amplified scenario demonstrates that when missingness patterns themselves are predictive, all mechanisms can achieve strong performance.

3.2.4 Error rate analysis

Type I and Type II error patterns are detailed in Appendix C. Key findings: MIMIC showed 3-4 fold Type I inflation above nominal 5% threshold across all conditions, while AMI showed more modest 1.4-1.7 fold inflation due to favorable dimensionality. Type II error varied dramatically by mechanism: MAR produced 50-100% false negative rates (true variables not detected), while MCAR and MNAR enabled near-perfect true variable identification. Trade-off between Type I and Type II errors was particularly stark under MAR, where false positive control resulted in near-complete signal detection failure.

3.2.5 Validation of real data findings

Simulation results validate six key patterns observed in real clinical data:

1. Missing indicators improve performance. Real data showed consistent benefits (Tables 4, 5; Figures 1, 2). Simulation confirms this under MNAR, where indicators doubled precision for AMI (Table 7), suggesting real data contains MNAR elements.

Table 8 Sensitivity Analysis: Prediction Performance When Outcomes Depend Solely on Missingness Patterns ($\alpha = 10$ Amplification)

Dataset	Mechanism	Method	AUC	F1 Score	ALPP
AMI	MAR	KNN	0.873 (0.035)	0.858 (0.049)	-0.217 (0.051)
		Mean	**0.883 (0.037)**	0.857 (0.047)	-0.215 (0.048)
		MICE	0.882 (0.039)	**0.860 (0.052)**	**-0.212 (0.054)**
		missForest	0.881 (0.038)	0.859 (0.052)	**-0.212 (0.055)**
	MCAR	KNN	0.943 (0.062)	0.912 (0.118)	-0.131 (0.098)
		Mean	**0.967 (0.027)**	0.940 (0.034)	-0.113 (0.047)
		MICE	0.954 (0.033)	0.933 (0.042)	-0.122 (0.057)
		missForest	0.966 (0.027)	**0.941 (0.031)**	**-0.107 (0.042)**
	MNAR	KNN	0.883 (0.036)	**0.862 (0.049)**	**-0.213 (0.053)**
		Mean	0.880 (0.042)	0.858 (0.049)	**-0.213 (0.055)**
		MICE	0.878 (0.037)	0.861 (0.051)	**-0.213 (0.053)**
		missForest	**0.889 (0.046)**	0.859 (0.044)	-0.214 (0.051)
MIMIC	MAR	KNN	0.956 (0.039)	0.813 (0.185)	-0.235 (0.079)
		Mean	0.940 (0.042)	0.823 (0.118)	-0.257 (0.095)
		MICE	**0.989 (0.015)**	**0.898 (0.108)**	**-0.118 (0.097)**
		missForest	0.983 (0.023)	0.862 (0.163)	-0.185 (0.168)
	MCAR	KNN	0.978 (0.024)	0.850 (0.089)	**-0.144 (0.073)**
		Mean	0.979 (0.030)	0.829 (0.110)	-0.159 (0.121)
		MICE	**0.981 (0.023)**	**0.875 (0.079)**	-0.151 (0.102)
		missForest	0.975 (0.038)	0.853 (0.098)	-0.159 (0.121)
	MNAR	KNN	0.913 (0.058)	0.816 (0.143)	-0.271 (0.126)
		Mean	0.937 (0.052)	0.814 (0.105)	-0.315 (0.196)
		MICE	0.972 (0.029)	0.842 (0.120)	-0.210 (0.115)
		missForest	**0.995 (0.010)**	**0.954 (0.054)**	**-0.086 (0.057)**

Predictor coefficients set to zero ($\boldsymbol{\beta}^{\text{true},X} = \mathbf{0}$), missing indicator coefficients amplified to $\boldsymbol{\beta}^{\text{true},Z} = \{1.0, 5.0, 10.0, 15.0\}$ (10-fold increase from base coefficients $\{0.1, 0.5, 1.0, 1.5\}$). This extreme MNAR scenario tests performance when missingness itself is the predictive signal. Compare with Tables 6 and 7 where outcomes depend on predictor values. Values presented as mean with standard deviation in parentheses across 10-fold cross-validation. Bold values indicate best performance for each metric within each dataset-mechanism combination.

Abbreviations: See Table 6. $\alpha = 10$: amplification factor.

2. Mean imputation achieves good prediction. Real data showed mean imputation performing comparably to sophisticated methods (Tables 4, 5). Simulation confirms this for prediction: for the AMI dataset under MCAR, mean imputation achieved strong predictive performance, only marginally worse than MICE and missForest (Table 7). This aligns with benchmarks showing mean with indicators is "never statistically significantly outperformed" [44] and "effective yet remarkably efficient" [45]. However, mean produces extreme coefficient inflation (400-fold higher bias under certain mechanisms, Table 6), making it unsuitable for inference [26].

3. Parsimonious models suffice. Real data's predictive performance peaked within 3-7 variables. With only 4 true variables in simulation, additional selections beyond this range are predominantly false positives. Type I analysis confirms substantial spurious selection, but model averaging in BAS and BART weights smaller models more heavily, limiting the impact of false positives on prediction.

4. High-dimensionality affects Type I error. MIMIC-III ($n/p = 3.2$) showed 2.4-fold higher Type I error than AMI ($n/p = 14.0$), and greater coefficient variability. Unfavorable ratios amplify false discovery and parameter instability.

5. MICE excels for inference. While mean imputation achieves competitive prediction, simulation reveals coefficient inflation makes it unsuitable for inference. MICE maintains more stable coefficient recovery under MCAR (Table 7), supporting its necessity when unbiased estimates and valid standard errors are required [7, 23].

6. Mechanism matters more than method. MAR caused large drops in predictive performance across all methods, demonstrating that no imputation approach can fully compensate when missingness depends on observed values. MCAR and MNAR enabled strong performance with appropriate method-indicator combinations. This validates the importance of understanding missingness mechanisms when selecting analysis strategies.

4 Discussion

BAS models selected small subsets (3-7 variables) across both datasets, supporting parsimonious clinical models. Missing indicators improved performance across most settings, with benefits most pronounced under high missingness. Mean imputation showed substantial benefits from using missing indicators in MIMIC-III, while MICE's conditional modeling partially captured missingness information. BART achieved better performance than BAS, while BAS provided clearer variable prioritization through distinct posterior inclusion probabilities (Tables 2, 3).

4.1 Interpretation of simulation findings

Simulation provided three critical insights. First, missingness mechanism determines achievable performance independent of method. Under MAR, substantial performance degradation across all methods demonstrates that no imputation approach can fully compensate when missingness depends on observed values. Second, elevated Type I and Type II errors do not preclude reliable prediction when BAS and tree ensembles BART compensate for spurious selections, though such violations remain problematic for statistical inference. Third, missing indicators provide benefits only when missingness itself carries predictive information (MNAR), explaining their variable effectiveness across mechanisms and their consistent utility in real clinical data. This was confirmed by sensitivity analysis where outcomes depended solely on missingness patterns (Table 8).

4.2 Implications for method selection

Our combined simulation and real data analysis provided the first systematic evaluation of imputation methods within a Bayesian variable selection framework, revealing that method appropriateness depends on analytical goals.

While prior work has examined imputation methods with frequentist variable selection [44–46], our study extended these findings to Bayesian approaches (BAS and BART), which offer distinct advantages for clinical applications: posterior inclusion probabilities for variable ranking and natural uncertainty quantification. Our results demonstrated that mean imputation with missing indicators achieves strong predictive performance in this Bayesian context. Mean imputation matched or exceeded sophisticated methods in our real data analysis (Tables 4, 5) and maintained strong discrimination despite Type I inflation in simulation, consistent with recent benchmarks in frequentist settings [44–46].

Critically, our comprehensive simulation under controlled missingness mechanisms (MCAR, MAR, MNAR) revealed a novel insight: the effectiveness of missing indicators depends fundamentally on the missingness mechanism itself. Under MNAR, where missingness patterns carry information about outcomes, indicators provide substantial benefits. Under MCAR, where missingness is uninformative, indicators offer minimal advantage. Under MAR, no method recovers lost information. This mechanism-dependent performance has important implications for method selection but cannot be determined from real data where true mechanisms are unknown.

However, our simulation also revealed that mean imputation's predictive success comes at a cost: severe coefficient inflation (Table 6) and elevated Type I errors make it unsuitable for statistical inference. MICE maintained stable coefficient recovery, supporting established principles that multiple imputation is necessary for unbiased inference [7, 23, 47].

These findings provided novel empirical evidence for a key theoretical distinction [26, 27]: method requirements differ fundamentally between risk stratification (where mean imputation with indicators provides efficiency and competitive performance) and effect estimation (where multiple imputation remains essential). Our contribution lies not in discovering this distinction, but in demonstrating it empirically within Bayesian variable selection and providing quantitative evidence across realistic clinical datasets and controlled simulation conditions.

While we cannot definitively determine missingness mechanisms in real data, our simulation-informed recommendations prioritize robustness: mean imputation with indicators performs well under both MCAR and MNAR, while MICE provides valid inference regardless of mechanism. When selecting among approaches, we recommend comparing multiple methods on held-out data to empirically assess performance for your specific dataset.

4.3 Context-specific recommendations

4.3.1 For clinical prediction and risk stratification

Recommended: Mean imputation + missing indicators + BART

This maximizes predictive accuracy with computational efficiency. Include indicators particularly when missingness exceeds 30%. Appropriate for ICU triage, emergency decision support, population health stratification, and clinical decision systems prioritizing accuracy. When predictive performance is primary and coefficient interpretation unnecessary, mean with indicators provides effective solutions. Achieving strong predictive performance on both datasets, the simulation's elevated Type I errors do not preclude prediction applications in Bayesian variable selection frameworks.

4.3.2 For inference and interpretable clinical models

Recommended: MICE + BAS (interpretability) or BART (flexibility)

Multiple imputation is essential for effect estimation and hypothesis testing. MICE maintained stable coefficient recovery and better error control (Tables 6, 7), providing proper uncertainty quantification for valid inference [7, 23]. Mean imputation should not be used for these applications due to severe coefficient inflation and elevated Type I errors that violate fundamental assumptions (controlled error rates, valid confidence intervals, unbiased estimates).

For interpretable models: BAS + MICE/missForest + missing indicators

BAS provides clear ranking through posterior inclusion probabilities. Appropriate for clinical decision rules, simple scoring systems, situations requiring transparent models for clinical adoption, and identifying critical predictors. This balances performance with transparency for clinical adoption.

4.4 Practical considerations

Dataset characteristics influence performance. AMI (6.15% missingness, $n/p = 14.0$) showed smaller indicator benefits and less Type I inflation (1.4-1.7 fold) versus MIMIC-III (64% missingness, $n/p = 3.2$, 3-4 fold inflation). Missing indicators provide greatest benefit in high-dimensional settings with substantial missingness.

Computational efficiency favors simpler approaches. Mean imputation requires minimal computation and straightforward implementation, while MICE requires specification, diagnostics, and longer runtime. For real-time applications, efficiency may justify mean imputation for prediction.

Our simulation demonstrated that MAR caused substantial performance degradation across all imputation methods for both datasets. This empirical result confirms a fundamental principle established in the missing data literature [7]: no amount of statistical sophistication can fully recover information that is fundamentally unavailable due to the missingness mechanism. This finding highlights the critical importance of study design that minimizes missingness rather than relying solely on post-hoc statistical corrections.

4.5 Limitations and future directions

We evaluated two datasets with binary outcomes and specific missingness patterns; results may not generalize to all domains, continuous outcomes, or survival analyses. Our primary analyses focused on selecting up to 20 variables, though we provide supplementary analyses using full variable sets (Appendices D, E) and alternative simulation configurations (Appendix F) for completeness.Simulation assumed specific generation processes that may not capture real data complexity. We cannot determine real data mechanisms (MCAR, MAR, MNAR), which prevents us from prescribing the optimal method. MIMIC-III's small sample ($n = 168$) in high dimensions ($p = 52$ to 103) represents a challenging scenario; results may differ with larger samples. AMI's low missingness (6.15%) provided limited opportunity to observe indicator benefits.

We did not perform external validation on independent datasets from different healthcare systems or geographic populations, nor did we assess temporal stability of our findings. Performance on new clinical sites, patient populations, or time periods remains unknown. As we did in this study, we recommend that practitioners validate their chosen method on held-out data and use simulation designs similar to ours to assess robustness under different missingness scenarios.

Future work should evaluate these approaches across diverse clinical domains with varying missingness patterns and outcome types. Developing principled methods for missingness mechanism diagnosis would enable more targeted recommendations. Investigating interactions between variable selection algorithms and imputation strategies beyond those examined here could reveal additional opportunities for improvement. Finally, extending this framework to time-varying missingness in longitudinal studies and competing risks in survival analyses are important future directions.

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Declarations

The primary author managed the entire experimental process, wrote the abstract, and provided an overview of the study. He also worked closely with the contributing author on the formulation of the study's objectives and scope. He received assistance from the contributing author in carrying out the experiments. The contributing author was instrumental in providing insightful feedback for the final manuscript's improvement and proofreading.

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- Conflict of interest/Competing interests
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- Ethics approval and consent to participate
  Not applicable.
- Consent for publication
  Not applicable.

- Data availability
  The MIMIC-III dataset is publicly available through PhysioNet (https://physionet.org/content/mimiciii/) following completion of required training and data use agreements. The AMI dataset is available from the corresponding author upon reasonable request.
- Materials availability
  Not applicable.
- Code availability
  All analysis code is currently being prepared for public release on GitHub and will be made available upon publication. In the interim, code is available from the corresponding author upon reasonable request.
- Author contribution
  Nads A.: Conceptualization, Methodology, Formal Analysis, Software, Writing–Original Draft, Visualization. Andrade D.: Conceptualization, Supervision, Writing–Review & Editing.

Appendix A Mathematical definitions of missingness mechanisms

Missing Completely at Random (MCAR): Missingness occurs independently of all variables:

$$P(R_{ij} = 1) = p_{\text{miss}}, \tag{A1}$$

where $R_{ij} = 1$ indicates X_{ij} is missing and p_{miss} is a constant probability calibrated to match the observed missingness rate in each dataset (64% for MIMIC-III, 6.15% for AMI).

Missing at Random (MAR): Missingness depends on observed variables:

$$P(R_{ij} = 1 \mid X) = \text{expit}(\alpha_0 + \alpha_1 X_{ik}), \quad k \neq j, \tag{A2}$$

where $\text{expit}(z) = 1/(1 + e^{-z})$ is the inverse logit function.

Missing Not at Random (MNAR): Missingness depends on the value itself:

$$P(R_{ij} = 1 \mid X) = \text{expit}(\alpha_0 + \alpha_1 X_{ij}). \tag{A3}$$

Parameters α_0 and α_1 were calibrated to achieve target missingness rates matching the real datasets.

Appendix B Implementation details and hyperparameters

Table B1 provides complete implementation details for all Bayesian variable selection methods and imputation approaches we used in this study.

Table B1 Implementation details and hyperparameters for all methods

Method	Parameter	Setting
BAS	MCMC iterations	1,500
	Prior	Robust g-prior with $g = n$ (90% of training data)
	Model prior	Beta-Binomial(1,1)
	Method	MCMC sampling
	R package	BAS [48]
BART	Number of trees (M)	200
	Burn-in iterations	100
	Post-burn-in samples	1,000
	Tree parameters	$k = 2.0$, power= 2.0, base= 0.95
	Function	lbart (logistic BART)
	Sparsity prior	Disabled
	R package	BART [49]
MICE	Number of imputations (m)	20 (real data), 3 (simulation)
	Iterations	1,000
	Method	PMM (Predictive Mean Matching)
	R package	mice [50]
missForest	Maximum iterations	1,000
	Number of trees	100
	R package	missForest [51]
KNN	Number of neighbors (k)	$\sqrt{n}$ MIMIC-III: $k \approx 13$; AMI: $k \approx 41$
	R package	VIM [52]
Mean imputation	Method	Column means for continuous variables
Cross-validation	Folds	10-fold
	Random seed	123 (for reproducibility)

Appendix C Type I and Type II error analysis

For the MIMIC-III dataset (Table C2), Type I error substantially exceeded the nominal 5% threshold across all conditions, inflating by approximately 3-4 fold. Mean imputation without indicators under MNAR achieved the lowest false positive rate, while missForest with indicators under MAR showed the highest inflation. Type II error varied widely, ranging from near-perfect identification under MNAR to substantial misses under MAR, where methods failed to identify approximately half to three-quarters of true variables.

The error pattern revealed a fundamental selection trade-off. Methods achieving lower Type I error tended toward higher Type II error, particularly under MAR and MNAR. Mean imputation under MNAR showed the most favorable balance, achieving low Type II error while maintaining relatively controlled Type I inflation.

For the AMI dataset (Table C3), Type I error showed more modest inflation (1.4-1.7 fold). The lower dimensional stress ($n/p = 14.0$ vs 3.2) enabled better false positive control. However, mechanism type dramatically affected Type II error: under MAR,

Table C2 Type I and Type II Error Results for MIMIC-III Dataset: False Positive and False Negative Rates Across Imputation Methods and Missingness Mechanisms

Mechanism	Method	mi	Type I Error (%)	Type II Error (%)
MAR	KNN	w/o mi	**17.98 (0.80)**	**45.00 (19.72)**
		w/ mi	18.59 (0.85)	60.00 (21.08)
	Mean	w/o mi	18.28 (1.00)	52.50 (24.86)
		w/ mi	18.69 (0.98)	62.50 (24.30)
	MICE	w/o mi	18.28 (1.00)	52.50 (24.86)
		w/ mi	19.39 (1.04)	80.00 (25.82)
	missForest	w/o mi	19.19 (1.17)	75.00 (28.87)
		w/ mi	19.49 (1.26)	82.50 (31.29)
MCAR	KNN	w/o mi	17.78 (0.71)	40.00 (17.48)
		w/ mi	18.18 (0.48)	50.00 (11.79)
	Mean	w/o mi	18.08 (0.32)	47.50 (7.91)
		w/ mi	18.18 (0.82)	50.00 (20.41)
	MICE	w/o mi	17.98 (0.80)	45.00 (19.72)
		w/ mi	18.08 (0.88)	47.50 (21.89)
	missForest	w/o mi	**17.47 (0.96)**	**32.50 (23.72)**
		w/ mi	17.78 (1.59)	40.00 (39.44)
MNAR	KNN	w/o mi	17.47 (1.07)	32.50 (26.48)
		w/ mi	18.18 (1.26)	50.00 (31.18)
	Mean	w/o mi	**16.57 (0.52)**	**10.00 (12.91)**
		w/ mi	18.18 (0.67)	50.00 (16.67)
	MICE	w/o mi	17.47 (1.17)	32.50 (28.99)
		w/ mi	18.59 (1.28)	60.00 (31.62)
	missForest	w/o mi	17.37 (1.24)	30.00 (30.73)
		w/ mi	18.28 (1.21)	52.50 (29.93)

Type I and Type II Error rates for four imputation methods (MICE, MEAN, missForest, KNN) with and without missing indicators (w/o mi, w/ mi) across three missingness mechanisms (MCAR, MAR, MNAR) for the MIMIC-III dataset. Type I Error represents the percentage of false positives in variable selection. Type II Error represents the percentage of false negatives (true variables not selected). Values presented as mean with standard deviation in parentheses across 10 folds. **Bold values indicate best performance for each metric within each mechanism–mi combination.**

Abbreviations: mi, missing indicator; MCAR, Missing Completely At Random; MAR, Missing At Random; MNAR, Missing Not At Random; w/ mi, with missing indicators; w/o mi, without missing indicators.

Type II error approached 100%; under MCAR and MNAR, several methods achieved perfect or near-perfect identification.

Under MCAR, MICE with missing indicators achieved optimal balance, eliminating false negatives while maintaining modest Type I inflation. Under MNAR, mean and missForest without indicators achieved perfect true variable identification with low false positive rates.

Mechanism type showed minimal effect on Type I error but dominated Type II patterns. MAR consistently produced worst Type II error, reflecting fundamental information loss. MCAR and MNAR enabled substantially better identification, though at elevated false positive rates.

Table C3 Type I and Type II Error Results for AMI Dataset: False Positive and False Negative Rates Across Imputation Methods and Missingness Mechanisms

Mechanism	Method	mi	Type I Error (%)	Type II Error (%)
MAR	KNN	w/o mi	8.35 (0.27)	95.00 (15.81)
		w/ mi	8.35 (0.18)	95.00 (10.54)
	Mean	w/o mi	8.35 (0.27)	95.00 (15.81)
		w/ mi	8.35 (0.18)	95.00 (10.54)
	MICE	w/o mi	8.44 (0.00)	100.00 (0.00)
		w/ mi	**8.31 (0.28)**	**92.50 (16.87)**
	missForest	w/o mi	8.35 (0.27)	95.00 (15.81)
		w/ mi	8.40 (0.13)	97.50 (7.91)
MCAR	KNN	w/o mi	6.84 (0.18)	5.00 (10.54)
		w/ mi	7.09 (0.27)	20.00 (15.81)
	Mean	w/o mi	6.96 (0.22)	12.50 (13.18)
		w/ mi	6.96 (0.30)	12.50 (17.68)
	MICE	w/o mi	6.88 (0.20)	7.50 (12.08)
		w/ mi	**6.75 (0.00)**	**0.00 (0.00)**
	missForest	w/o mi	6.92 (0.22)	10.00 (12.91)
		w/ mi	7.09 (0.27)	20.00 (15.81)
MNAR	KNN	w/o mi	6.88 (0.20)	7.50 (12.08)
		w/ mi	6.79 (0.13)	2.50 (7.91)
	Mean	w/o mi	**6.75 (0.00)**	**0.00 (0.00)**
		w/ mi	6.88 (0.20)	7.50 (12.08)
	MICE	w/o mi	6.79 (0.13)	2.50 (7.91)
		w/ mi	6.79 (0.13)	2.50 (7.91)
	missForest	w/o mi	**6.75 (0.00)**	**0.00 (0.00)**
		w/ mi	6.79 (0.13)	2.50 (7.91)

Type I and Type II Error rates for four imputation methods (MICE, MEAN, missForest, KNN) with and without missing indicators (w/o mi, w/ mi) across three missingness mechanisms (MCAR, MAR, MNAR) for the AMI dataset. Type I Error represents the percentage of false positives in variable selection. Type II Error represents the percentage of false negatives (true variables not selected). Values presented as mean with standard deviation in parentheses across 10 folds. **Bold values indicate best performance for each metric within each mechanism–MI combination.**

Abbreviations: mi, missing indicator; MCAR, Missing Completely At Random; MAR, Missing At Random; MNAR, Missing Not At Random; w/ mi, with missing indicators; w/o mi, without missing indicators.

Appendix D Model performance using full variable set on MIMIC-III data

Figure D1 presents the average log predicted probability for BAS (top row) and BART (bottom row) applied to the full MIMIC-III dataset using increasing percentages (10% to 100%) of the available variables across three experimental settings. The results demonstrate a consistent pattern: predictive performance typically plateaus or even declines beyond the first 20–30% of variables across all settings.

This plateau effect has important implications for clinical decision-making and model interpretability. Including all available variables does not improve predictive performance and often introduces unnecessary complexity that can hinder clinical

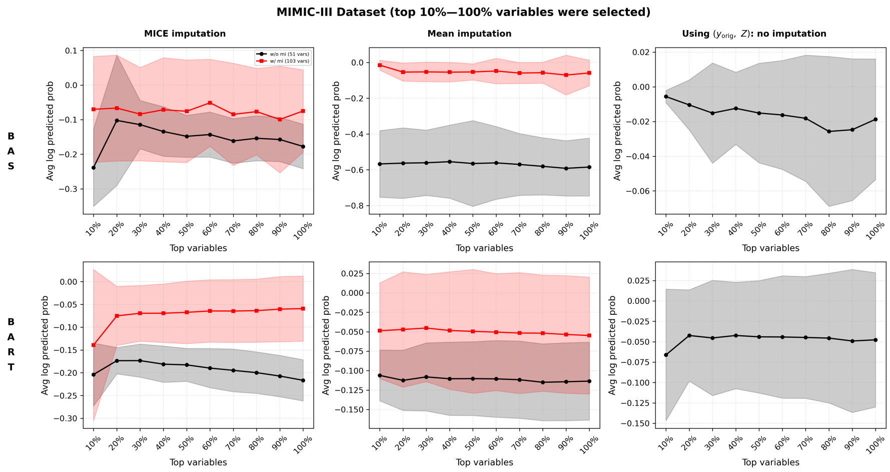

Fig. D1 Average log predicted probability from BAS (top row) and BART (bottom row) applied to the original MIMIC-III dataset using all available variables. Each column shows results under a different condition: MICE imputation (left), mean imputation (middle), and no imputation using only missing indicators Z (right). Variable subsets are selected based on cumulative percentages from 10% to 100%. Shaded regions represent standard deviation across folds.

interpretation. The performance curves show that a carefully selected subset of variables captures the essential predictive information, while additional variables primarily contribute noise rather than meaningful signal. This finding is particularly relevant for clinical applications where model transparency and feature interpretability are crucial for physician acceptance and regulatory approval.

The consistency of this pattern across both BAS and BART methods, and across different data preparation approaches (MICE, mean imputation, and missing indicator analysis), suggests that the phenomenon is robust and not dependent on specific modeling assumptions. From a clinical decision-making perspective, models using a smaller set of well-selected variables are not only equally effective but also more practical for implementation, as they require fewer data inputs and provide clearer insights into which clinical factors drive predictions. This supports the value of principled variable selection in developing deployable clinical prediction models.

Appendix E Model performance using full variable set on AMI data

Figure E2 presents the average log predicted probabilities for BAS (top row) and BART (bottom row) on the AMI dataset using increasing percentages of variables. The results demonstrate an even more pronounced pattern of performance degradation compared to the MIMIC-III dataset, with predictive performance consistently declining as more variables are included across all three experimental settings.

In the MICE imputation setting (left column), both BAS and BART show clear deterioration in performance as the variable set expands beyond the initial 10–20%. The mean imputation setting (center column) exhibits similar patterns, with optimal

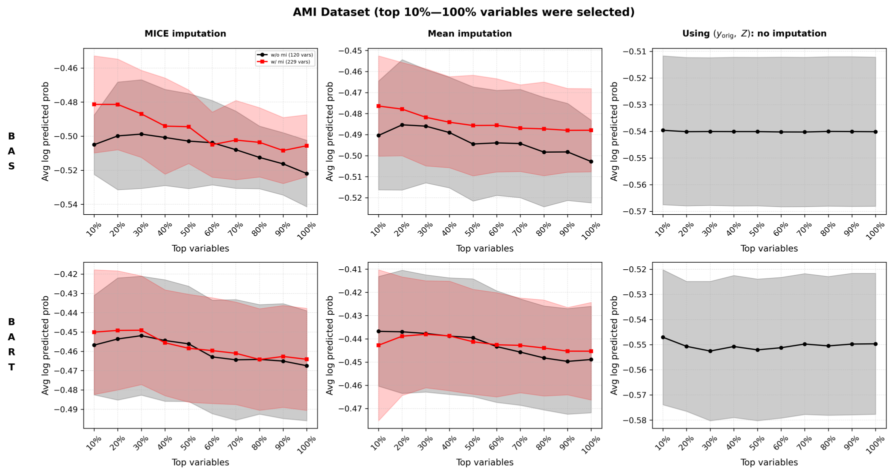

Fig. E2 Average log predicted probability from BAS (top row) and BART (bottom row) applied to the original AMI dataset using all available variables. Each column shows results under a different condition: MICE imputation (left), mean imputation (middle), and no imputation using only missing indicators Z (right). Variable subsets are selected based on cumulative percentages from 10% to 100%. Shaded regions represent standard deviation across folds.

performance achieved using only the top-ranked variables, followed by steady decline when additional features are incorporated. Notably, the missing indicators-only setting Z (right column) shows relatively stable but consistently lower performance compared to the imputed data approaches, suggesting that the actual covariate values provide more predictive information than missingness patterns alone for this clinical outcome.

These findings are particularly striking for the AMI dataset, as they demonstrate that the curse of dimensionality becomes more severe with larger feature spaces (229 variables with missing indicators vs. 103 in MIMIC-III). The consistent downward trend across both Bayesian methods reinforces that including weakly informative variables not only fails to improve predictions but actively degrades model performance. This validates the critical importance of principled variable selection in high-dimensional clinical prediction tasks, where the signal-to-noise ratio decreases substantially as irrelevant features accumulate.

Appendix F Simulation implementation details

F.1 Signal strength calculation

Signal strength values $c \in \{1, 2, 3\}$ were applied to selected variables. For missing indicator simulations, c was assigned directly to the 5 variables with highest missing rates (30–70% missing).

F.2 Model selection criteria

GLMNET [53] models were retained only when the number of nonzero coefficients satisfied $4 \leq \#\{j : \hat{\boldsymbol{\beta}}_{\lambda,j} \neq 0\} \leq 8$. Fixed intercept $\alpha = 0.5$ was used across all simulations.

F.3 Expected log-likelihood computation

For missing indicator simulations:

$$\mathbb{E}[\log p(y|z_i)] = p(y=1|z_i)\log p(y=1|z_i) + p(y=0|z_i)\log p(y=0|z_i) \tag{F4}$$

where $p(y=1|z_i) = \sigma(z_i^\top \beta^{\text{true}})$.

F.4 Supplementary figure: BAS and BART performance on simulated MIMIC-III data using missing indicators

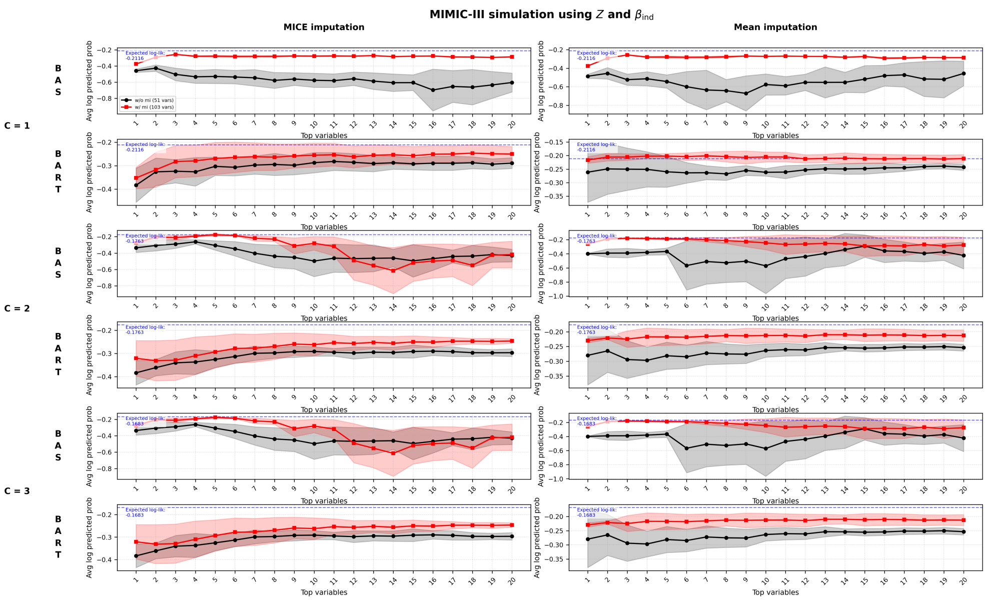

Fig. F3 Average log predicted probability from BAS and BART applied to simulated data from the MIMIC-III dataset. The predictor matrix was simulated as $X \sim \mathcal{N}(0,1)$, and the binary response variable y was generated using the missing indicator matrix Z and indicator coefficient vector $\boldsymbol{\beta}_{\text{ind}}$ under varying signal strengths $c \in \{1,2,3\}$. The left column shows results from MICE-imputed data; the right column shows results from mean-imputed data. Shaded regions indicate standard deviation across folds. Comparisons are made between models with and without missing indicators.

Figure F3 summarizes the performance of BAS and BART models under the simulation approach based on the MIMIC-III dataset. Across all signal strength values

$c \in \{1, 2, 3\}$, models that include missing indicators (red lines) generally achieve higher average log predicted probabilities than those without (black lines). This improvement is particularly pronounced with mean imputation (right column), where the inclusion of missing indicators consistently leads to better predictive performance across both methods and all signal strengths.

Under MICE imputation (left column), the performance gap between models with and without missing indicators is considerably narrower, especially for BART. This suggests that MICE imputation already captures much of the latent structure in the data, thereby reducing the incremental value provided by explicit missing indicators. The capacity of BART to model nonlinear interactions further minimizes this gap, as it can extract meaningful patterns even from imputed data alone.

The horizontal dashed lines in each plot represent the expected log-likelihood under the true generative process, serving as theoretical upper bounds for predictive performance. As signal strength c increases from 1 to 3, both BAS and BART models demonstrate progressively improved performance, confirming that the signal embedded in the missingness patterns is being successfully captured. Notably, the models approach but do not exceed these theoretical limits, indicating realistic performance expectations.

These results emphasize the differential value of missing indicators across imputation strategies: they provide substantial benefit under simpler imputation methods (mean) but offer diminishing returns when sophisticated imputation (MICE) is employed. The findings reinforce the importance of considering both the underlying signal strength and the choice of imputation method when evaluating the utility of missing indicators in predictive modeling frameworks.

F.5 Supplementary figure: BAS and BART performance on simulated AMI data using missing indicators

Figure F4 shows the results for AMI dataset-based simulations under varying signal strengths $c \in \{1, 2, 3\}$. The results demonstrate even more pronounced benefits of missing indicators compared to the MIMIC-III simulations, reflecting the higher dimensionality and missingness patterns in the AMI dataset. As signal strength c increases, both BAS and BART models show substantially improved average log predicted probabilities when missing indicators are included (red lines) compared to models without them (black lines).

This pattern is particularly striking under mean imputation (right column), where BAS models with missing indicators rapidly approach the expected log-likelihood (horizontal dashed lines) even at moderate signal strengths. The close alignment between model performance and theoretical benchmarks indicates that the models are effectively recovering the underlying signal encoded in the missingness patterns Z, especially when combined with simpler imputation methods that leave more informational gaps for the missing indicators to fill.

Under MICE imputation (left column), the benefits of missing indicators remain substantial but are somewhat attenuated, consistent with the MIMIC-III findings. BART models show similar patterns of improvement with missing indicators, though the gains are more modest than those observed with BAS. This suggests that while

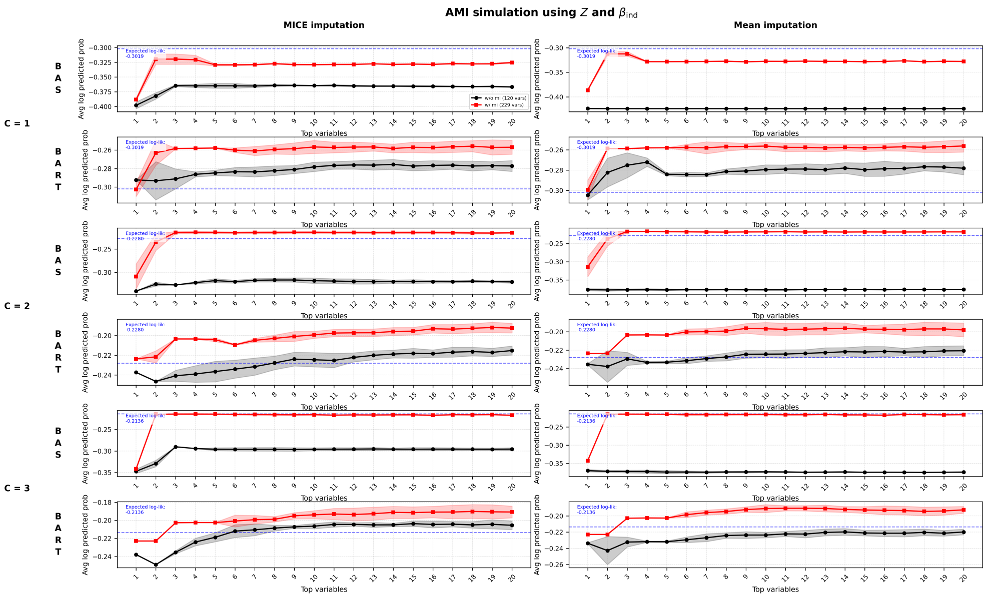

Fig. F4 Average log predicted probability from BAS and BART applied to simulated data from the AMI dataset. The predictor matrix was simulated as $X \sim \mathcal{N}(0, 1)$, and the binary response variable y was generated using the missing indicator matrix Z and the indicator coefficient vector $\boldsymbol{\beta}_{\text{ind}}$ under varying signal strengths $c \in \{1, 2, 3\}$. The left column displays the results using MICE-imputed data, while the right column displays the results using mean-imputed data. Shaded regions represent standard deviation across folds. Each model is evaluated with and without the inclusion of missing indicators.

BART's capacity for modeling complex nonlinear relationships provides some compensation for missing information, explicit missing indicators still contribute meaningful predictive value, particularly as signal strength increases.

The AMI dataset results reinforce that missing indicators provide the greatest utility when: (1) the underlying missingness signal is strong, (2) simpler imputation methods are employed, and (3) linear modeling approaches (BAS) are used that may not capture complex patterns as effectively as tree-based methods (BART). These findings underscore the importance of considering dataset characteristics and modeling choices when evaluating the incremental value of missing indicators in high-dimensional clinical prediction tasks.

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